

#1 · Basic Physiology Anatomy Pharmacology and Genetics

Which ion channel and current is responsible for phase 0 of the ventricular myocyte action potential?

Answer

The fast voltage-gated sodium channel Nav1.5 (encoded by SCN5A) generates I_{Na} , producing the rapid upstroke of phase 0.

#2 · Basic Physiology Anatomy Pharmacology and Genetics

What ionic currents dominate phase 1 of the ventricular action potential?

Answer

The transient outward potassium current I_{to} , carried mainly by Kv4.3, produces the brief early repolarization notch of phase 1.

#3 · Basic Physiology Anatomy Pharmacology and Genetics

What two opposing currents are balanced during phase 2 (the plateau) of the ventricular action potential?

Answer

Inward calcium current through L-type calcium channels (I_{CaL}) is balanced against outward delayed rectifier potassium currents (I_{Kr} and I_{Ks}), maintaining the plateau.

#4 · Basic Physiology Anatomy Pharmacology and Genetics

Which two delayed rectifier potassium currents drive phase 3 repolarization, and what genes encode them?

Answer

I_{Kr} (encoded by $KCNH2$, also called hERG) and I_{Ks} (encoded by $KCNQ1$ with its beta subunit $KCNE1$) drive phase 3 repolarization.

#5 · Basic Physiology Anatomy Pharmacology and Genetics

What maintains phase 4 resting membrane potential in a ventricular myocyte?

Answer

The inward rectifier potassium current I_{K1} , carried by $Kir2.1$, holds the resting membrane potential near the potassium equilibrium potential of about negative 90 mV.

#6 · Basic Physiology Anatomy Pharmacology and Genetics

What ionic current is uniquely responsible for phase 4 diastolic depolarization in SA nodal cells?

Answer

The funny current I_f , carried by HCN channels (mainly $HCN4$) permeable to sodium and potassium, produces slow spontaneous phase 4 depolarization in pacemaker cells.

#7 · Basic Physiology Anatomy Pharmacology and Genetics

Why does the SA node action potential upstroke depend on calcium rather than sodium current?

Answer

SA nodal cells have a less negative resting potential that inactivates fast sodium channels, so the upstroke (phase 0) is generated by the slow inward L-type calcium current instead of I_{Na} .

#8 · Basic Physiology Anatomy Pharmacology and Genetics

What is the "calcium clock" mechanism of SA nodal automaticity?

Answer

Spontaneous rhythmic sarcoplasmic reticulum calcium release via ryanodine receptors activates the sodium-calcium exchanger (NCX), generating an inward depolarizing current that contributes to diastolic depolarization independent of the membrane clock.

#9 · Basic Physiology Anatomy Pharmacology and Genetics

How does the sodium-calcium exchanger (NCX) affect membrane potential during a normal cardiac cycle?

Answer

NCX exchanges 3 sodium ions in for 1 calcium ion out, making it electrogenic and net depolarizing (inward current) when extruding calcium during diastole.

#10 · Basic Physiology Anatomy Pharmacology and Genetics

What structural protein forms the channels of a gap junction, and what determines conduction anisotropy?

Answer

Connexins (mainly connexin-43 in ventricular myocardium) form gap junction channels; their higher density at intercalated disc ends versus lateral borders creates faster longitudinal than transverse conduction (anisotropy).

#11 · Basic Physiology Anatomy Pharmacology and Genetics

Which connexin predominates in the working atrial and ventricular myocardium versus the specialized conduction system?

Answer

Connexin-43 predominates in atrial and ventricular myocardium, while connexin-40 predominates in the atria and His-Purkinje system, contributing to their faster conduction velocity.

#12 · Basic Physiology Anatomy Pharmacology and Genetics

Where is the sinoatrial (SA) node anatomically located?

Answer

The SA node lies subepicardially at the junction of the right atrial appendage and superior vena cava, near the crista terminalis.

#13 · Basic Physiology Anatomy Pharmacology and Genetics

Where is the AV node located within Koch's triangle?

Answer

The compact AV node sits at the apex of Koch's triangle, bordered by the tendon of Todaro, the septal tricuspid valve leaflet, and the coronary sinus OS.

#14 · Basic Physiology Anatomy Pharmacology and Genetics

Why is AV nodal conduction so slow compared to atrial or His-Purkinje tissue?

Answer

AV nodal cells have calcium-dependent slow-response action potentials with decremental conduction and few gap junctions, producing conduction velocity of only about 0.02 to 0.05 m/s, which allows the delay needed for ventricular filling.

#15 · Basic Physiology Anatomy Pharmacology and Genetics

What is the approximate conduction velocity of the His-Purkinje system, and why is it so fast?

Answer

His-Purkinje conduction velocity is about 2 to 4 m/s, the fastest in the heart, due to large cell diameter, abundant gap junctions, and high sodium channel density.

#16 · Basic Physiology Anatomy Pharmacology and Genetics

What structure is the anatomic target for pulmonary vein isolation in atrial fibrillation ablation?

Answer

The antral region surrounding the ostia of the four pulmonary veins is targeted, since sleeves of atrial myocardium extending into the pulmonary veins are the dominant trigger source for paroxysmal atrial fibrillation.

#17 · Basic Physiology Anatomy Pharmacology and Genetics

What is the cavotricuspid isthmus, and why is it clinically important?

Answer

The cavotricuspid isthmus is the region of right atrial tissue between the inferior vena cava and the tricuspid annulus; it is the critical zone of slow conduction targeted for ablation of typical (cavotricuspid isthmus-dependent) atrial flutter.

#18 · Basic Physiology Anatomy Pharmacology and Genetics

Where does the coronary sinus drain, and why is it important for EP procedures?

Answer

The coronary sinus drains venous blood into the right atrium near the tricuspid annulus; it provides catheter access to the left atrial and ventricular epicardium and houses the musculature connecting to the left atrium relevant to some atrial arrhythmias.

#19 · Basic Physiology Anatomy Pharmacology and Genetics

What defines the effective refractory period (ERP) of cardiac tissue?

Answer

The ERP is the longest S1-S2 coupling interval at which a premature stimulus fails to produce a propagated action potential, reflecting the time needed for enough sodium channels to recover from inactivation.

#20 · Basic Physiology Anatomy Pharmacology and Genetics

What is the difference between the relative refractory period and the effective refractory period?

Answer

During the relative refractory period a stimulus can still propagate but produces a slower, smaller-amplitude response, while during the effective refractory period no propagated response is possible at all.

#21 · Basic Physiology Anatomy Pharmacology and Genetics

Why does the AV node normally have a longer effective refractory period than atrial or ventricular tissue?

Answer

The AV node's calcium-dependent slow-response cells have prolonged recovery of excitability, giving it the longest refractory period of the conduction axis and allowing it to protect the ventricles from rapid atrial rates.

#22 · Basic Physiology Anatomy Pharmacology and Genetics

What effect does vagal (parasympathetic) stimulation have on sinus rate and AV nodal conduction?

Answer

Vagal stimulation releases acetylcholine acting on M2 receptors, activating IKACH to hyperpolarize and slow SA nodal automaticity and slow or block AV nodal conduction by prolonging its refractory period.

#23 · Basic Physiology Anatomy Pharmacology and Genetics

What is the cellular mechanism by which acetylcholine slows the sinus node?

Answer

Acetylcholine binds M2 muscarinic receptors coupled to Gi proteins, which activate the acetylcholine-sensitive potassium current I_{KACH} and inhibit adenylate cyclase, reducing I_f and I_{CaL} , thereby slowing phase 4 depolarization.

#24 · Basic Physiology Anatomy Pharmacology and Genetics

How does sympathetic stimulation increase heart rate at the cellular level?

Answer

Norepinephrine acting on beta-1 adrenergic receptors activates Gs proteins and adenylate cyclase, increasing cAMP, which enhances I_f and I_{CaL} to accelerate phase 4 diastolic depolarization and increase sinus rate.

#25 · Basic Physiology Anatomy Pharmacology and Genetics

What is decremental conduction, and where is it most prominent?

Answer

Decremental conduction is progressive slowing and eventual failure of impulse propagation as conduction velocity decreases with each successive beat; it is most prominent in the AV node.

#26 · Basic Physiology Anatomy Pharmacology and Genetics

What defines concealed conduction?

Answer

Concealed conduction occurs when an impulse penetrates part of the conduction system (often the AV node) without producing a visible propagated beat, but alters the refractoriness or timing of subsequent impulses.

#27 · Basic Physiology Anatomy Pharmacology and Genetics

What is the classic electrophysiologic hallmark distinguishing enhanced automaticity from reentry as an arrhythmia mechanism?

Answer

Enhanced automaticity typically cannot be reproducibly initiated or terminated by programmed electrical stimulation and shows a warm-up phenomenon, whereas reentry can typically be reproducibly initiated and terminated with pacing.

#28 · Basic Physiology Anatomy Pharmacology and Genetics

What are the three classic requirements for a reentrant circuit to form and sustain?

Answer

Reentry requires two functionally or anatomically distinct conduction pathways, unidirectional block in one pathway, and slow conduction allowing the other pathway time to recover excitability before the impulse returns.

#29 · Basic Physiology Anatomy Pharmacology and Genetics

What is the difference between early afterdepolarizations (EADs) and delayed afterdepolarizations (DADs)?

Answer

EADs occur during phase 2 or 3 of the action potential (before full repolarization) and are associated with prolonged QT and bradycardia, while DADs occur after full repolarization (phase 4) and are associated with calcium overload states like digoxin toxicity or catecholaminergic stress.

#30 · Basic Physiology Anatomy Pharmacology and Genetics

What ionic mechanism underlies early afterdepolarizations?

Answer

Prolonged action potential duration allows recovery from inactivation and reactivation of L-type calcium channels during phase 2 or 3, producing a secondary inward current that triggers an afterdepolarization.

#31 · Basic Physiology Anatomy Pharmacology and Genetics

What ionic mechanism underlies delayed afterdepolarizations?

Answer

Spontaneous sarcoplasmic reticulum calcium release during diastole activates the sodium-calcium exchanger in its forward mode, generating a transient inward depolarizing current after full repolarization.

#32 · Basic Physiology Anatomy Pharmacology and Genetics

What is triggered activity, and how does it differ from automaticity?

Answer

Triggered activity is an arrhythmia arising from afterdepolarizations that occur only in response to a preceding action potential, unlike automaticity, which arises spontaneously from phase 4 depolarization without requiring a preceding beat.

#33 · Basic Physiology Anatomy Pharmacology and Genetics

What is the mechanism of action of Vaughan Williams class Ia antiarrhythmics such as quinidine and procainamide?

Answer

Class Ia drugs block fast sodium channels with intermediate dissociation kinetics, slowing phase 0 upstroke and conduction velocity while also prolonging the action potential duration and QT interval via potassium channel blockade.

#34 · Basic Physiology Anatomy Pharmacology and Genetics

What distinguishes class Ib antiarrhythmics like lidocaine and mexiletine from class Ia and Ic?

Answer

Class Ib drugs have fast sodium channel dissociation kinetics, preferentially block channels in ischemic or depolarized tissue, shorten action potential duration slightly, and have minimal effect on the QT interval.

#35 · Basic Physiology Anatomy Pharmacology and Genetics

What distinguishes class Ic antiarrhythmics like flecainide and propafenone from class Ia and Ib?

Answer

Class Ic drugs have slow dissociation kinetics from sodium channels, producing marked slowing of conduction velocity (widened QRS) with little effect on action potential duration or repolarization.

#36 · Basic Physiology Anatomy Pharmacology and Genetics

What is the mechanism of Vaughan Williams class II antiarrhythmics?

Answer

Class II drugs are beta-adrenergic receptor blockers that reduce sympathetic-driven automaticity, slow AV nodal conduction, and prolong AV nodal refractoriness by reducing cAMP-mediated calcium and funny current activity.

#37 · Basic Physiology Anatomy Pharmacology and Genetics

What is the primary electrophysiologic effect of Vaughan Williams class III drugs such as amiodarone and sotalol?

Answer

Class III drugs primarily block potassium channels (especially I_{Kr}) to prolong action potential duration and the effective refractory period, which prolongs the QT interval.

#38 · Basic Physiology Anatomy Pharmacology and Genetics

Why is amiodarone considered to have multi-class antiarrhythmic activity despite being classified as class III?

Answer

Amiodarone blocks potassium, sodium, and calcium channels and also has noncompetitive beta-blocking activity, giving it properties of classes I, II, III, and IV simultaneously.

#39 · Basic Physiology Anatomy Pharmacology and Genetics

What is the mechanism of Vaughan Williams class IV antiarrhythmics such as verapamil and diltiazem?

Answer

Class IV drugs are non-dihydropyridine calcium channel blockers that inhibit L-type calcium channels, slowing conduction and prolonging refractoriness predominantly in the SA and AV nodes.

#40 · Basic Physiology Anatomy Pharmacology and Genetics

Why does amiodarone require a long loading period and have an extremely long elimination half-life?

Answer

Amiodarone is highly lipophilic and extensively distributed into adipose and other tissues, producing a large volume of distribution and an elimination half-life of roughly 40 to 60 days after chronic dosing.

#41 · Basic Physiology Anatomy Pharmacology and Genetics

Why must dofetilide dosing be adjusted for renal function and initiated with inpatient QT monitoring?

Answer

Dofetilide is eliminated predominantly by renal tubular secretion, so reduced creatinine clearance raises plasma levels and increases risk of excessive QT prolongation and torsades de pointes, requiring dose adjustment and telemetry monitoring at initiation.

#42 · Basic Physiology Anatomy Pharmacology and Genetics

What is the mechanism by which adenosine terminates AV nodal reentrant tachycardia?

Answer

Adenosine activates A1 receptors coupled to Gi proteins, opening IKAdo potassium channels and inhibiting cAMP-mediated calcium current, producing transient hyperpolarization and AV nodal block that interrupts the reentrant circuit.

#43 · Basic Physiology Anatomy Pharmacology and Genetics

Which gene encodes the cardiac sodium channel implicated in type 3 long QT syndrome and Brugada syndrome?

Answer

SCN5A encodes the alpha subunit of the cardiac sodium channel Nav1.5; gain-of-function mutations cause long QT syndrome type 3, while loss-of-function mutations cause Brugada syndrome.

#44 · Basic Physiology Anatomy Pharmacology and Genetics

Which gene and current are responsible for long QT syndrome type 1, and what is its classic clinical trigger?

Answer

KCNQ1 encodes the IKs channel alpha subunit; loss-of-function mutations cause long QT syndrome type 1, classically triggered by exercise or swimming.

#45 · Basic Physiology Anatomy Pharmacology and Genetics

Which gene and current are responsible for long QT syndrome type 2, and what is its classic clinical trigger?

Answer

KCNH2 (hERG) encodes the IKr channel alpha subunit; loss-of-function mutations cause long QT syndrome type 2, classically triggered by sudden loud noise or emotional stress.

#46 · Basic Physiology Anatomy Pharmacology and Genetics

Which gene is most commonly mutated in catecholaminergic polymorphic ventricular tachycardia (CPVT)?

Answer

RYR2, encoding the cardiac ryanodine receptor, is the most common gene mutated in autosomal dominant CPVT, causing abnormal sarcoplasmic reticulum calcium leak that triggers delayed afterdepolarizations during catecholamine surges.

#47 · Basic Physiology Anatomy Pharmacology and Genetics

What genetic inheritance pattern and gene defect cause Jervell and Lange-Nielsen syndrome?

Answer

Jervell and Lange-Nielsen syndrome is an autosomal recessive form of long QT syndrome caused by biallelic mutations in KCNQ1 (or KCNE1), associated with severe QT prolongation and congenital sensorineural deafness.

#48 · Basic Physiology Anatomy Pharmacology and Genetics

Which sarcomeric gene is most commonly mutated in hypertrophic cardiomyopathy?

Answer

MYH7, encoding beta-myosin heavy chain, and MYBPC3, encoding cardiac myosin-binding protein C, are the two most commonly implicated genes in hypertrophic cardiomyopathy.

#49 · Basic Physiology Anatomy Pharmacology and Genetics

Which gene is most frequently associated with arrhythmogenic right ventricular cardiomyopathy (ARVC)?

Answer

PKP2, encoding the desmosomal protein plakophilin-2, is the most commonly mutated gene in arrhythmogenic right ventricular cardiomyopathy.

#50 · Basic Physiology Anatomy Pharmacology and Genetics

What is the inheritance pattern of most channelopathies causing long QT syndrome, Brugada syndrome, and CPVT?

Answer

Most channelopathies including long QT syndrome types 1 and 2 and Brugada syndrome are inherited in an autosomal dominant pattern, while CPVT is most commonly autosomal dominant (RYR2) though an autosomal recessive form (CASQ2) also exists.

#51 · Clinical Arrhythmias Core Concepts

What is 60-cycle (AC) artifact on an intracardiac or surface ECG recording and how does it typically appear?

Answer

A fine, regular oscillation at 60 Hz caused by nearby electrical equipment or poor grounding. It appears as a uniform, high-frequency sinusoidal noise superimposed on the baseline that does not vary with cardiac cycle length.

#52 · Clinical Arrhythmias Core Concepts

How can electrocautery artifact during a device procedure mimic ventricular fibrillation on telemetry?

Answer

Electrocautery produces high-frequency, high-amplitude chaotic signal that saturates the amplifier and looks similar to VF. Correlating with the patient's clinical status, arterial line waveform, or pulse oximeter tracing helps confirm the patient is not actually in VF.

#53 · Clinical Arrhythmias Core Concepts

What is a common cause of artifact mimicking atrial fibrillation on a surface ECG in a tremulous patient?

Answer

Skeletal muscle tremor (such as Parkinsonian tremor) creates irregular, low-amplitude baseline oscillations that can mimic fibrillatory waves, but the QRS complexes remain regularly spaced, distinguishing it from true atrial fibrillation.

#54 · Clinical Arrhythmias Core Concepts

What ECG feature helps differentiate lead misplacement artifact from a true arrhythmia?

Answer

Lead misplacement often produces an abrupt, isolated change in QRS morphology or axis on a single beat or lead without any change in the underlying rhythm, and repeating the ECG with corrected lead placement resolves the finding.

#55 · Clinical Arrhythmias Core Concepts

What is electromagnetic interference (EMI) oversensing in a pacemaker or ICD, and why is it clinically important?

Answer

EMI oversensing occurs when the device misinterprets external electrical noise as intrinsic cardiac signals, which can cause inappropriate inhibition of pacing or inappropriate ICD shocks; sources include electrocautery, MRI, and some industrial equipment.

#56 · Clinical Arrhythmias Core Concepts

What is a three-dimensional electroanatomic mapping system used for in the EP lab?

Answer

It creates a real-time, three-dimensional geometric and electrical map of a cardiac chamber by tracking catheter position and local electrograms, allowing the operator to identify arrhythmia substrate and guide ablation with reduced fluoroscopy use.

#57 · Clinical Arrhythmias Core Concepts

What is the difference between contact mapping and non-contact mapping systems?

Answer

Contact mapping requires the catheter tip to physically touch the endocardium to record each point, while non-contact mapping uses a multi-electrode array in the chamber to reconstruct thousands of virtual electrograms across the chamber surface without direct contact at every site.

#58 · Clinical Arrhythmias Core Concepts

What is high-density mapping in electroanatomic systems, and what advantage does it provide?

Answer

High-density mapping uses multi-electrode catheters that acquire many closely spaced points per catheter position, producing more detailed voltage and activation maps that better characterize scar borders and complex circuits than sparse point-by-point mapping.

#59 · Clinical Arrhythmias Core Concepts

How is artificial intelligence being incorporated into modern EP mapping systems?

Answer

AI-assisted mapping algorithms can automatically annotate local activation times, filter and classify far-field versus near-field signals, identify likely scar or conduction channels, and suggest optimal ablation targets, reducing operator-dependent variability in map interpretation.

#60 · Clinical Arrhythmias Core Concepts

What is a key limitation of AI-based automated activation annotation in electroanatomic mapping?

Answer

Automated algorithms can mis-annotate activation timing in areas of fractionated or double potentials, low voltage scar, or significant noise, so the operator must still manually review and correct annotations rather than accept them without verification.

#61 · Clinical Arrhythmias Core Concepts

What is ripple mapping, and how does it differ from conventional isochronal activation mapping?

Answer

Ripple mapping displays every local electrogram deflection as a moving bar without requiring manual annotation of activation time, visually representing wavefront propagation dynamically, which can better reveal complex or discontinuous conduction than a fixed color-coded isochronal map.

#62 · Clinical Arrhythmias Core Concepts

What is the purpose of tilt-table testing, and what constitutes a positive result for vasovagal syncope?

Answer

Tilt-table testing evaluates susceptibility to neurally mediated syncope by passively tilting the patient upright while monitoring blood pressure and heart rate. A positive test reproduces symptoms along with hypotension, bradycardia, or both corresponding to the patient's typical presyncope or syncope.

#63 · Clinical Arrhythmias Core Concepts

What are the three hemodynamic response subtypes seen on tilt-table testing?

Answer

The cardioinhibitory type shows significant bradycardia or asystole, the vasodepressor type shows significant hypotension without much heart rate change, and the mixed type shows both bradycardia and hypotension.

#64 · Clinical Arrhythmias Core Concepts

On a surface ECG, what QRS duration threshold generally defines a wide QRS complex?

Answer

A QRS duration of 120 milliseconds or greater in adults is generally considered wide and prompts evaluation for bundle branch block, ventricular pacing, preexcitation, or ventricular origin of the rhythm.

#65 · Clinical Arrhythmias Core Concepts

What morphologic features on a wide complex tachycardia favor ventricular tachycardia over supraventricular tachycardia with aberrancy?

Answer

AV dissociation, fusion or capture beats, extreme left axis deviation, a QRS duration greater than 160 milliseconds, and concordance of QRS polarity across the precordial leads all favor ventricular tachycardia.

#66 · Clinical Arrhythmias Core Concepts

What is the Brugada algorithm used for in wide complex tachycardia interpretation?

Answer

It is a stepwise ECG algorithm that evaluates absence of RS complex in precordial leads, RS interval duration, AV dissociation, and morphology criteria to help distinguish ventricular tachycardia from supraventricular tachycardia with aberrant conduction.

#67 · Clinical Arrhythmias Core Concepts

What is the main advantage of an implantable loop recorder compared with a 30-day external event monitor?

Answer

An implantable loop recorder provides continuous monitoring for up to several years, which allows detection of infrequent arrhythmic events such as syncope or atrial fibrillation that occur less often than can reliably be captured by a short-term external monitor.

#68 · Clinical Arrhythmias Core Concepts

How do consumer smartwatch single-lead ECG and photoplethysmography features contribute to arrhythmia detection, and what is a key limitation?

Answer

They can flag irregular pulse patterns suggestive of atrial fibrillation and allow the patient to record a single-lead tracing during symptoms, but they have lower diagnostic accuracy than a 12-lead ECG or physician-interpreted monitor and can generate false positive alerts requiring clinical confirmation.

#69 · Clinical Arrhythmias Core Concepts

What is the basic goal of programmed electrical stimulation during an invasive electrophysiology study?

Answer

Programmed electrical stimulation delivers timed extrastimuli and burst pacing from intracardiac catheters to assess conduction properties, refractory periods, and to attempt induction of clinical arrhythmias for diagnosis and mapping.

#70 · Clinical Arrhythmias Core Concepts

What does the AH interval measure during an EP study, and what is its typical normal range?

Answer

The AH interval measures conduction time through the AV node, from the low right atrial electrogram to the His bundle deflection, and is normally about 60 to 130 milliseconds; it is autonomically sensitive and can lengthen with vagal tone or AV nodal disease.

#71 · Clinical Arrhythmias Core Concepts

What does the HV interval measure, and what value suggests infra-Hisian disease?

Answer

The HV interval measures conduction from the His bundle to the onset of ventricular activation through the His-Purkinje system, normally 35 to 55 milliseconds; a value greater than 55 to 70 milliseconds suggests infra-Hisian conduction disease with higher risk of progression to complete heart block.

#72 · Clinical Arrhythmias Core Concepts

What is the difference between the effective refractory period and the functional refractory period in EP testing?

Answer

The effective refractory period is the longest coupling interval of a premature stimulus that fails to propagate through a tissue, while the functional refractory period is the shortest interval between two consecutive depolarizations that emerge from that tissue.

#73 · Clinical Arrhythmias Core Concepts

What is the biophysical mechanism by which radiofrequency ablation destroys myocardial tissue?

Answer

Radiofrequency energy passes alternating current through tissue, generating resistive (Joule) heating at the electrode-tissue interface that produces coagulation necrosis when tissue temperature exceeds approximately 50 degrees Celsius.

#74 · Clinical Arrhythmias Core Concepts

What is a steam pop during radiofrequency ablation, and why is it dangerous?

Answer

A steam pop is an audible or visible explosive release of steam caused by rapid overheating of tissue at the catheter tip, which can cause a cardiac perforation or crater lesion and is a marker of excessive power delivery or poor catheter-tissue contact.

#75 · Clinical Arrhythmias Core Concepts

How does irrigated-tip radiofrequency ablation reduce the risk of char and thrombus formation compared with non-irrigated catheters?

Answer

Irrigated catheters continuously flush saline through pores at the catheter tip to cool the electrode surface, which limits coagulum formation and char while allowing deeper lesion formation at higher power without excessive surface temperature.

#76 · Clinical Arrhythmias Core Concepts

What is the biophysical mechanism of tissue injury during cryoablation?

Answer

Cryoablation circulates a refrigerant such as nitrous oxide through the catheter tip, cooling tissue to below negative 40 degrees Celsius, which causes cell death through ice crystal formation, cellular dehydration, and microvascular injury leading to apoptosis and coagulation necrosis.

#77 · Clinical Arrhythmias Core Concepts

What clinical advantage does cryoablation have over radiofrequency ablation when ablating near the AV node?

Answer

Cryoablation allows cryomapping, in which a reversible, lower-temperature test lesion can confirm the desired electrophysiologic effect before delivering a full freeze, and cryo lesions carry a lower risk of causing permanent AV block if terminated during a test freeze.

#78 · Clinical Arrhythmias Core Concepts

What is the mechanism of tissue injury in pulsed field ablation, and how does it differ fundamentally from thermal ablation?

Answer

Pulsed field ablation delivers brief high-voltage electrical pulses that create irreversible electroporation, forming permanent nanoscale pores in the cell membrane that lead to cell death, without relying on significant heating or cooling of the tissue.

#79 · Clinical Arrhythmias Core Concepts

Why is pulsed field ablation considered to have a favorable safety profile near structures such as the esophagus and phrenic nerve?

Answer

Cardiac myocytes have a lower electroporation threshold than esophageal, nerve, and vascular tissue, so pulsed field ablation can preferentially injure myocardium at energy levels that largely spare adjacent extracardiac structures, reducing risk of esophageal injury and phrenic nerve palsy.

#80 · Clinical Arrhythmias Core Concepts

What is a recognized biophysical limitation or safety concern specific to pulsed field ablation despite its tissue selectivity?

Answer

Pulsed field ablation can still cause coronary artery spasm, hemolysis from red blood cell electroporation with resultant acute kidney injury, and microbubble formation that requires careful catheter-tissue contact and energy delivery protocols to minimize gas embolism risk.

#81 · Clinical Arrhythmias Bradycardias

What is sinus node dysfunction (SND)?

Answer

A disorder of impaired sinus node impulse generation or conduction to the surrounding atrium, producing inappropriate sinus bradycardia, sinus pauses or arrest, sinoatrial exit block, or chronotropic incompetence.

#82 · Clinical Arrhythmias Bradycardias

What is tachycardia-bradycardia syndrome?

Answer

A form of sinus node dysfunction in which paroxysms of atrial tachyarrhythmia, most often atrial fibrillation, alternate with sinus bradycardia or long pauses on termination of the tachycardia, often producing presyncope at the moment of conversion.

#83 · Clinical Arrhythmias Bradycardias

What is chronotropic incompetence?

Answer

The inability of the sinus node to increase heart rate appropriately with exertion or physiologic demand, commonly defined as failure to reach 80 percent of age-predicted maximal heart rate during exercise testing, and a recognized cause of exertional fatigue and dyspnea.

#84 · Clinical Arrhythmias Bradycardias

What ECG finding distinguishes first-degree AV block?

Answer

A PR interval greater than 200 milliseconds with every P wave conducted to a QRS complex; there is delay in AV conduction but no dropped beats.

#85 · Clinical Arrhythmias Bradycardias

How does Mobitz type I (Wenckebach) second-degree AV block appear on ECG?

Answer

Progressive PR interval prolongation with each successive beat until a P wave fails to conduct and a QRS complex is dropped, after which the cycle resets; it typically localizes to the AV node.

#86 · Clinical Arrhythmias Bradycardias

How does Mobitz type II second-degree AV block appear on ECG?

Answer

A sudden, unpredictable dropped QRS complex without preceding PR interval prolongation, with constant PR intervals in the conducted beats; it typically reflects infranodal (His-Purkinje) disease and carries a higher risk of progression to complete heart block.

#87 · Clinical Arrhythmias Bradycardias

What defines third-degree (complete) AV block?

Answer

Complete AV dissociation in which no atrial impulses conduct to the ventricles, so P waves and QRS complexes occur independently at their own regular rates, with ventricular activity maintained by a junctional or ventricular escape rhythm.

#88 · Clinical Arrhythmias Bradycardias

How can escape rhythm morphology help localize the block level in complete heart block?

Answer

A narrow QRS escape rhythm at 40 to 60 beats per minute suggests a junctional escape focus with block at or above the AV node, while a wide QRS escape rhythm at 20 to 40 beats per minute suggests a ventricular escape focus with infranodal block, which is less stable and more likely to require urgent pacing.

#89 · Clinical Arrhythmias Bradycardias

Why does 2:1 AV block require additional evaluation to localize the block site?

Answer

Because every other P wave is blocked, there is no sequence of PR prolongation to observe, so the level (nodal versus infranodal) cannot be determined from the resting ECG alone; exercise, carotid sinus massage, vagal maneuvers, or His bundle electrogram recording during electrophysiology study can help distinguish the two.

#90 · Clinical Arrhythmias Bradycardias

How does exercise typically affect nodal versus infranodal AV block, and why is this useful?

Answer

Exercise (via increased sympathetic tone) typically improves conduction and narrows the block ratio in nodal (AV node) block but worsens conduction and increases the block ratio in infranodal (His-Purkinje) block, helping localize the site of 2:1 or high-grade block.

#91 · Clinical Arrhythmias Bradycardias

Is permanent pacing indicated for asymptomatic Mobitz I second-degree AV block localized to the AV node?

Answer

Generally no; asymptomatic Mobitz I block confined to the AV node has a benign prognosis and low risk of progression, so permanent pacing is not routinely indicated unless the patient is symptomatic or the block is shown to be infranodal.

#92 · Clinical Arrhythmias Bradycardias

Is permanent pacing indicated for asymptomatic Mobitz II or high-grade AV block?

Answer

Yes; per ACC/AHA/HRS guidelines, permanent pacing is a Class I indication for Mobitz II, high-grade, or third-degree AV block even in asymptomatic patients, because of the substantial risk of progression to symptomatic complete heart block or asystole.

#93 · Clinical Arrhythmias Bradycardias

What is the pacing recommendation for symptomatic sinus node dysfunction?

Answer

Permanent pacing is a Class I indication when bradycardia, pauses, or chronotropic incompetence from sinus node dysfunction are clearly correlated with symptoms such as syncope, presyncope, or fatigue that cannot be attributed to a reversible cause.

#94 · Clinical Arrhythmias Bradycardias

When is permanent pacing indicated for reflex or neurally mediated bradycardia?

Answer

Pacing is reasonable in patients with recurrent, unpredictable syncope due to documented symptomatic pauses from carotid sinus hypersensitivity or a cardioinhibitory vasovagal response, particularly in older patients where a reversible or situational trigger has been excluded.

#95 · Clinical Arrhythmias Bradycardias

What is conduction system pacing?

Answer

A pacing strategy that places the lead to directly capture the native His-Purkinje conduction system, either at the His bundle or in the left bundle branch area, in order to achieve near-physiologic ventricular activation rather than the dyssynchronous activation produced by conventional right ventricular septal or apical pacing.

#96 · Clinical Arrhythmias Bradycardias

What is His bundle pacing and what is its main electrical advantage?

Answer

His bundle pacing places a fixed-screw lead directly into or adjacent to the His bundle so the paced impulse travels down the native conduction system, typically producing a narrow paced QRS complex that closely resembles the patient's intrinsic QRS and preserves physiologic ventricular synchrony.

#97 · Clinical Arrhythmias Bradycardias

What are the main practical limitations of His bundle pacing?

Answer

His bundle pacing is technically demanding to achieve reliable fixation, often has higher and less stable capture thresholds that can rise over time, produces lower sensed R-wave amplitudes, and carries a meaningfully higher long-term lead revision rate than left bundle branch area pacing.

#98 · Clinical Arrhythmias Bradycardias

What is left bundle branch area pacing (LBBAP)?

Answer

A conduction system pacing technique in which the lead is screwed deep into the interventricular septum until it captures the left bundle branch or adjacent conducting fascicles, producing rapid left ventricular activation with a characteristic right bundle branch block pattern paced QRS morphology.

#99 · Clinical Arrhythmias Bradycardias

Why has left bundle branch area pacing become more widely adopted than His bundle pacing?

Answer

LBBAP generally achieves lower and more stable long-term capture thresholds, higher sensed R-wave amplitudes, and greater implant success and lead stability than His bundle pacing, while still providing much more physiologic ventricular activation than conventional right ventricular pacing.

#100 · Clinical Arrhythmias Bradycardias

What ECG and electrogram findings confirm successful left bundle branch capture during implant?

Answer

A right bundle branch block-pattern paced QRS with a short and stable left ventricular activation time measured to the peak of the R wave in lead V6, along with demonstration of selective or nonselective left bundle branch capture criteria such as a discrete left bundle potential and an abrupt shortening of the stimulus-to-V6 R-wave peak time at threshold.

#101 • Clinical Arrhythmias Bradycardias

In which pacing population is conduction system pacing most strongly favored over conventional right ventricular pacing?

Answer

Patients anticipated to require a high burden of ventricular pacing, such as those with AV block and preserved or borderline left ventricular function, since conduction system pacing avoids the pacing-induced cardiomyopathy risk associated with sustained right ventricular apical or septal pacing.

#102 • Clinical Arrhythmias Bradycardias

What is pacing-induced cardiomyopathy and how does conduction system pacing address it?

Answer

It is a decline in left ventricular ejection fraction caused by chronic dyssynchronous ventricular activation from conventional right ventricular pacing; because conduction system pacing engages the native His-Purkinje network to produce synchronous activation, it substantially reduces this risk compared with right ventricular pacing.

#103 · Clinical Arrhythmias Bradycardias

What is a leadless pacemaker?

Answer

A self-contained pacing device implanted entirely within the heart, most commonly the right ventricle, via a catheter delivery system through the femoral vein, eliminating the transvenous lead and subcutaneous pulse generator pocket used in conventional pacemakers.

#104 · Clinical Arrhythmias Bradycardias

What are the main advantages of leadless pacemakers over transvenous systems?

Answer

They eliminate lead-related complications such as lead fracture, dislodgement, and venous occlusion, and eliminate pocket-related complications such as infection, hematoma, and erosion, since there is no lead and no subcutaneous device pocket.

#105 • Clinical Arrhythmias Bradycardias

What patient populations are especially well suited to single-chamber leadless pacemakers?

Answer

Patients with limited or difficult venous access, prior device infection requiring an extraction-avoidant strategy, hemodialysis-dependent patients who need their veins preserved for future access, and patients needing only infrequent ventricular pacing such as for permanent atrial fibrillation with slow ventricular response.

#106 • Clinical Arrhythmias Bradycardias

How does a dual-chamber leadless pacemaker system achieve AV synchronous pacing without an atrial lead?

Answer

Two separate leadless devices, one implanted in the right atrium and one in the right ventricle, communicate beat to beat using implant-to-implant conducted electrical communication through the body tissue, allowing the ventricular device to sense atrial activity and deliver AV synchronous ventricular pacing.

#107 · Clinical Arrhythmias Bradycardias

Why was the development of dual-chamber leadless pacing clinically significant?

Answer

Earlier leadless pacemakers were limited to single-chamber right ventricular pacing without true AV synchrony, so extending leadless technology to reliable dual-chamber AV synchronous pacing meaningfully broadened the pool of patients, including those with sinus node dysfunction or AV block needing atrial-based pacing, who could avoid a transvenous system entirely.

#108 · Clinical Arrhythmias Bradycardias

What is a recognized limitation of current leadless pacemaker technology compared with transvenous systems?

Answer

Leadless pacemakers generally cannot provide cardiac resynchronization therapy, have a more limited battery longevity and retrieval experience for very long-term follow-up, and device retrieval after long-term fibrous encapsulation can be more technically challenging than removing a transvenous lead.

#109 • Clinical Arrhythmias Bradycardias

Can a leadless pacemaker be combined with a subcutaneous implantable cardioverter-defibrillator?

Answer

Yes; a leadless pacemaker can be implanted alongside a subcutaneous ICD to provide antibradycardia and, in newer communicating systems, antitachycardia pacing support, giving patients who need both defibrillation and pacing an entirely transvenous-lead-free option.

#110 • Clinical Arrhythmias Bradycardias

Why is escape rhythm reliability a key clinical concern in acute complete heart block, regardless of the eventual pacing strategy chosen?

Answer

Because the ventricular escape rhythm in complete heart block can be slow and unstable with a risk of prolonged asystole, patients require continuous monitoring and often temporary pacing support until a permanent pacing solution, whether conventional, conduction system, or leadless, can be implanted.

#111 · Clinical Arrhythmias Atrial

What is the dominant mechanism sustaining most clinical atrial fibrillation?

Answer

Multiple wavelet reentry and rapidly firing ectopic foci, most often originating from myocardial sleeves within the pulmonary veins, interacting with an atrial substrate capable of sustaining fibrillatory conduction.

#112 · Clinical Arrhythmias Atrial

Why do the pulmonary veins serve as the dominant trigger source for atrial fibrillation?

Answer

Myocardial muscle sleeves extend from the left atrium into the pulmonary veins and have shorter, heterogeneous refractory periods and automaticity that allow rapid ectopic firing capable of initiating fibrillatory conduction.

#113 · Clinical Arrhythmias Atrial

How does atrial fibrosis promote AF perpetuation?

Answer

Fibrosis creates zones of slow, heterogeneous conduction that fractionate wavefronts and support multiple reentrant circuits, converting paroxysmal AF into a more persistent, self-sustaining arrhythmia over time.

#114 · Clinical Arrhythmias Atrial

What does the phrase AF begets AF describe?

Answer

Sustained atrial fibrillation itself causes electrical and structural remodeling, including shortened atrial refractoriness and progressive fibrosis, which makes the atrium increasingly likely to sustain future episodes.

#115 · Clinical Arrhythmias Atrial

What monitoring duration is generally preferred to detect paroxysmal AF after a cryptogenic stroke?

Answer

Prolonged monitoring, typically an implantable loop recorder providing weeks to years of continuous monitoring, substantially increases AF detection compared with short-term Holter monitoring of 24 to 48 hours.

#116 · Clinical Arrhythmias Atrial

What AF burden threshold on implantable monitors is commonly used to trigger consideration of anticoagulation in subclinical AF trials?

Answer

Device-detected atrial high-rate episodes lasting at least 5 to 6 minutes, though the optimal duration and rate threshold for anticoagulation decisions remain individualized based on stroke risk and episode burden.

#117 · Clinical Arrhythmias Atrial

How do wearable single-lead ECG patches and smartwatches contribute to AF screening?

Answer

They allow ambulatory, patient-triggered or continuous rhythm capture over days to weeks, increasing detection of asymptomatic paroxysmal AF compared with intermittent office-based ECG screening.

#118 · Clinical Arrhythmias Atrial

What is the first-line pharmacologic strategy for rate control in most patients with AF?

Answer

Beta-blockers or nondihydropyridine calcium channel blockers such as diltiazem or verapamil are first-line, targeting a resting heart rate generally below 110 beats per minute for a lenient rate-control strategy.

#119 · Clinical Arrhythmias Atrial

Why should nondihydropyridine calcium channel blockers be avoided for rate control in AF with reduced ejection fraction?

Answer

Diltiazem and verapamil have negative inotropic effects that can worsen heart failure with reduced ejection fraction, so beta-blockers or digoxin are preferred in that setting.

#120 · Clinical Arrhythmias Atrial

What is the role of digoxin in AF rate control?

Answer

Digoxin can help control resting heart rate, particularly in patients with heart failure, but it is less effective during exertion and is typically used as an add-on rather than first-line monotherapy.

#121 · Clinical Arrhythmias Atrial

What did the RACE II trial show about rate control targets in AF?

Answer

Lenient rate control, targeting resting heart rate below 110 beats per minute, was noninferior to strict rate control targeting below 80 beats per minute for a composite of cardiovascular outcomes.

#122 · Clinical Arrhythmias Atrial

Which antiarrhythmic drugs are considered first-line for rhythm control in AF without structural heart disease?

Answer

Flecainide and propafenone are first-line class IC options in patients without significant structural heart disease or ischemia, while amiodarone, dofetilide, and sotalol are used when structural disease is present.

#123 · Clinical Arrhythmias Atrial

Why must class IC antiarrhythmics like flecainide be combined with an AV nodal blocking agent when used for AF?

Answer

Flecainide and propafenone can slow atrial rate enough to organize AF into atrial flutter with 1:1 AV conduction, producing a very rapid ventricular rate, so an AV nodal blocker is co-administered to prevent this.

#124 · Clinical Arrhythmias Atrial

In what population is dofetilide preferred over other rhythm-control drugs, and what monitoring does it require?

Answer

Dofetilide is useful in patients with coronary disease or heart failure, and it requires inpatient initiation with continuous ECG monitoring and renal-function-based dosing due to risk of QT prolongation and torsades de pointes.

#125 · Clinical Arrhythmias Atrial

What is the mechanism and major toxicity profile concern of amiodarone when used long-term for AF?

Answer

Amiodarone blocks multiple ion channels including potassium, sodium, and calcium channels and has beta-blocking properties, but chronic use carries risk of thyroid dysfunction, pulmonary fibrosis, hepatotoxicity, and corneal deposits, so it is generally reserved for patients with structural heart disease or when other agents fail.

#126 · Clinical Arrhythmias Atrial

What did the EAST-AFNET 4 trial demonstrate about early rhythm control in AF?

Answer

Early rhythm control initiated within one year of AF diagnosis, using antiarrhythmic drugs or ablation, reduced cardiovascular death, stroke, and hospitalization for heart failure or acute coronary syndrome compared with usual care focused on rate control.

#127 · Clinical Arrhythmias Atrial

What is postoperative atrial fibrillation and when does it most commonly occur?

Answer

Postoperative AF is new-onset AF following cardiac or noncardiac surgery, most commonly occurring within the first 2 to 4 days after cardiac surgery due to inflammation, autonomic changes, and atrial stretch.

#128 · Clinical Arrhythmias Atrial

What is the evidence-based first-line pharmacologic prophylaxis against postoperative AF after cardiac surgery?

Answer

Perioperative beta-blockade, continued or initiated before surgery, is the most consistently effective and guideline-recommended strategy to reduce the incidence of postoperative AF.

#129 · Clinical Arrhythmias Atrial

Does postoperative AF after cardiac surgery require long-term anticoagulation in every patient?

Answer

Not necessarily; the decision should be individualized using standard stroke risk stratification such as CHA₂DS₂-VASc, since postoperative AF is associated with increased long-term stroke and AF recurrence risk even if it resolves before discharge.

#130 · Clinical Arrhythmias Atrial

What are the components of the CHA₂DS₂-VASc score?

Answer

Congestive heart failure, hypertension, age 75 or older (2 points), diabetes, prior stroke or TIA or thromboembolism (2 points), vascular disease, age 65 to 74, and female sex, used to estimate stroke risk in nonvalvular AF.

#131 • Clinical Arrhythmias Atrial

At what CHA₂DS₂-VASc score is oral anticoagulation generally recommended in men versus women with AF?

Answer

Anticoagulation is generally recommended for men with a score of 2 or higher and women with a score of 3 or higher, since female sex alone without other risk factors confers minimal independent stroke risk.

#132 • Clinical Arrhythmias Atrial

Why are direct oral anticoagulants generally preferred over warfarin for stroke prevention in nonvalvular AF?

Answer

DOACs such as apixaban, rivaroxaban, dabigatran, and edoxaban have shown noninferior or superior efficacy with lower rates of intracranial hemorrhage compared with warfarin, and they do not require routine INR monitoring.

#133 · Clinical Arrhythmias Atrial

In which patients with AF is warfarin still preferred over DOACs for stroke prevention?

Answer

Warfarin is preferred in patients with moderate to severe mitral stenosis or a mechanical heart valve, since DOACs have not shown efficacy and were associated with harm in these valvular AF populations.

#134 · Clinical Arrhythmias Atrial

What is the mechanism of action of dabigatran?

Answer

Dabigatran is a direct thrombin inhibitor that reversibly blocks the active site of thrombin, preventing fibrinogen conversion to fibrin.

#135 · Clinical Arrhythmias Atrial

What is the mechanism of action of apixaban and rivaroxaban?

Answer

Both are direct factor Xa inhibitors that block the conversion of prothrombin to thrombin by inhibiting free and clot-bound factor Xa.

#136 · Clinical Arrhythmias Atrial

What reversal agents are available for DOAC-associated life-threatening bleeding?

Answer

Idarucizumab reverses dabigatran, and andexanet alfa reverses factor Xa inhibitors such as apixaban and rivaroxaban.

#137 · Clinical Arrhythmias Atrial

What is the rationale for left atrial appendage occlusion devices in AF stroke prevention?

Answer

In nonvalvular AF, the left atrial appendage is the source of the majority of atrial thrombi, so mechanically excluding it from circulation can reduce stroke risk in patients who have a long-term contraindication to anticoagulation.

#138 · Clinical Arrhythmias Atrial

What device is most widely used for percutaneous left atrial appendage occlusion, and what trial supported its approval?

Answer

The Watchman device is most widely used, supported by the PROTECT AF and PREVAIL trials, which showed noninferiority to warfarin for stroke prevention with a reduction in hemorrhagic complications over long-term follow-up.

#139 · Clinical Arrhythmias Atrial

What antithrombotic regimen typically follows Watchman left atrial appendage occlusion implantation?

Answer

A short course of anticoagulation or dual antiplatelet therapy is typically continued for 45 days to allow device endothelialization, followed by transesophageal echocardiography assessment before transitioning to long-term antiplatelet therapy.

#140 · Clinical Arrhythmias Atrial

What is the minimum recommended duration of anticoagulation before elective cardioversion of AF lasting more than 48 hours or of unknown duration?

Answer

At least 3 weeks of therapeutic anticoagulation before cardioversion, followed by at least 4 weeks after, to reduce the risk of thromboembolism from atrial thrombus dislodgement.

#141 · Clinical Arrhythmias Atrial

How can transesophageal echocardiography shorten the pre-cardioversion anticoagulation requirement?

Answer

TEE can exclude left atrial or left atrial appendage thrombus, allowing cardioversion to proceed without the standard 3 weeks of prior anticoagulation, provided therapeutic anticoagulation is maintained at the time of and after the procedure.

#142 · Clinical Arrhythmias Atrial

What is the typical initial energy setting for synchronized electrical cardioversion of AF using a biphasic defibrillator?

Answer

An initial biphasic energy of 120 to 200 joules is typically used, with anterior-posterior pad placement often improving success compared with anterior-lateral placement.

#143 · Clinical Arrhythmias Atrial

Why must electrical cardioversion for AF be synchronized to the QRS complex?

Answer

Synchronization ensures the shock is delivered away from the vulnerable period of the T wave, avoiding induction of ventricular fibrillation.

#144 · Clinical Arrhythmias Atrial

What is the general target for pulmonary vein isolation during catheter ablation of AF?

Answer

Electrical isolation of all pulmonary veins from the left atrium, confirmed by demonstrating entrance and ideally exit block, since the pulmonary veins are the dominant trigger source for paroxysmal AF.

#145 · Clinical Arrhythmias Atrial

What did the CABANA and CASTLE-AF trials show about catheter ablation in AF?

Answer

CABANA showed ablation was not superior to drug therapy for the primary composite endpoint in the overall trial population but improved quality of life and reduced AF recurrence, while CASTLE-AF showed ablation reduced death and heart failure hospitalization in patients with AF and heart failure with reduced ejection fraction.

#146 · Clinical Arrhythmias Atrial

How does radiofrequency ablation achieve pulmonary vein isolation?

Answer

Thermal energy delivered through point-by-point or contact-force-guided catheter lesions creates a continuous circumferential scar around the pulmonary vein antra, electrically disconnecting the veins from the left atrium.

#147 · Clinical Arrhythmias Atrial

How does cryoballoon ablation achieve pulmonary vein isolation, and what is its main procedural advantage?

Answer

A balloon catheter delivers freezing energy circumferentially around the pulmonary vein ostium in a single application per vein, which simplifies the procedure compared with point-by-point radiofrequency ablation while achieving comparable outcomes for paroxysmal AF.

#148 · Clinical Arrhythmias Atrial

What is pulsed field ablation and what energy modality does it use?

Answer

Pulsed field ablation delivers high-voltage, short-duration electrical pulses that cause irreversible electroporation of cell membranes, rather than using thermal energy, to achieve tissue ablation.

#149 · Clinical Arrhythmias Atrial

What is the primary safety advantage of pulsed field ablation over thermal ablation modalities?

Answer

Pulsed field ablation exhibits relative tissue selectivity for myocardium at clinically used energy levels, reducing collateral thermal injury to adjacent structures such as the esophagus, phrenic nerve, and pulmonary veins compared with radiofrequency or cryoablation.

#150 · Clinical Arrhythmias Atrial

Why is esophageal injury less of a concern with pulsed field ablation compared with thermal ablation?

Answer

Because irreversible electroporation preferentially affects cardiomyocytes over other tissue types at standard energy settings, atrioesophageal fistula risk, a rare but often fatal complication of thermal ablation, is substantially reduced.

#151 · Clinical Arrhythmias Atrial

What did pivotal trials such as ADVENT show regarding pulsed field ablation compared with conventional thermal ablation for paroxysmal AF?

Answer

Pulsed field ablation demonstrated noninferior effectiveness for freedom from atrial arrhythmia recurrence and a comparable or favorable safety profile compared with radiofrequency or cryoballoon ablation.

#152 · Clinical Arrhythmias Atrial

What is a recognized mechanism-specific complication of pulsed field ablation related to unintended tissue effects?

Answer

Hemolysis from mechanical shear and electroporation of red blood cells can occur with high-energy pulsed field delivery, and coronary artery vasospasm has been reported when energy is delivered near coronary structures.

#153 · Clinical Arrhythmias Atrial

Why does pulsed field ablation typically shorten overall procedure time compared with radiofrequency ablation?

Answer

Each pulsed field application ablates tissue nearly instantaneously across the catheter contact area, and because thermal effects are minimal, less time is needed for lesion consolidation or cooling, reducing total ablation time per vein.

#154 · Clinical Arrhythmias Atrial

What additional ablation targets beyond pulmonary vein isolation may be considered in persistent or long-standing persistent AF?

Answer

Posterior wall isolation, ablation of complex fractionated atrial electrograms, linear lesions, and targeting of non-pulmonary vein triggers such as the superior vena cava or coronary sinus may be considered, though evidence for added benefit over pulmonary vein isolation alone is mixed.

#155 · Clinical Arrhythmias Atrial

What is the classic surgical Cox-Maze procedure designed to treat AF?

Answer

The Cox-Maze procedure creates a series of surgical incisions or ablation lines in both atria that compartmentalize the atrial tissue into small segments, interrupting the reentrant circuits and macro-reentry pathways that sustain AF while preserving a pathway for sinus impulse propagation.

#156 · Clinical Arrhythmias Atrial

In what clinical setting is the surgical Maze procedure most commonly performed today?

Answer

It is most commonly performed as a concomitant procedure during other open cardiac surgery, such as mitral valve surgery or coronary artery bypass grafting, in patients with a history of AF.

#157 · Clinical Arrhythmias Atrial

What is the purpose of left atrial appendage exclusion or excision when performed alongside surgical Maze or other cardiac surgery?

Answer

Surgical exclusion or excision of the left atrial appendage reduces the primary source of thrombus formation in AF, lowering long-term stroke risk independent of the rhythm outcome of the Maze lesion set.

#158 · Clinical Arrhythmias Atrial

When is AV junction ablation with permanent pacing considered for AF management?

Answer

It is considered in patients with symptomatic AF and poorly controlled ventricular rate despite optimal pharmacologic therapy, particularly when rhythm control is not feasible or desired, accepting a pace-and-ablate strategy rather than restoring sinus rhythm.

#159 · Clinical Arrhythmias Atrial

What is the fundamental limitation of AV junction ablation as an AF treatment strategy?

Answer

It renders the patient permanently pacemaker dependent without eliminating the underlying atrial fibrillation, so it controls ventricular rate rather than treating the atrial arrhythmia itself, and biventricular or conduction system pacing is often preferred to avoid pacing-induced cardiomyopathy from long-term right ventricular pacing.

#160 · Clinical Arrhythmias Atrial

What is the mechanism of typical, cavotricuspid isthmus-dependent atrial flutter?

Answer

A macro-reentrant circuit rotates around the tricuspid annulus, with the cavotricuspid isthmus, the region between the inferior vena cava and tricuspid valve, serving as a critical zone of slow conduction necessary to sustain the circuit.

#161 · Clinical Arrhythmias Atrial

What is the classic ECG appearance of typical counterclockwise cavotricuspid isthmus-dependent atrial flutter?

Answer

Negative sawtooth flutter waves in the inferior leads II, III, and aVF, and a positive flutter wave in lead V1, reflecting counterclockwise activation around the tricuspid annulus.

#162 · Clinical Arrhythmias Atrial

How does clockwise typical atrial flutter differ from counterclockwise typical flutter on ECG?

Answer

Clockwise typical flutter, which rotates in the opposite direction around the same cavotricuspid isthmus-dependent circuit, produces positive flutter waves in the inferior leads and a negative flutter wave in V1, the reverse of the counterclockwise pattern.

#163 · Clinical Arrhythmias Atrial

What is the typical atrial rate in cavotricuspid isthmus-dependent atrial flutter, and what ventricular rate commonly results?

Answer

The atrial rate is typically around 300 beats per minute, and with common 2:1 AV conduction the ventricular rate is approximately 150 beats per minute.

#164 · Clinical Arrhythmias Atrial

What defines atypical atrial flutter, in contrast to typical flutter?

Answer

Atypical atrial flutter is a macro-reentrant atrial tachycardia that does not depend on the cavotricuspid isthmus, instead using other anatomic or scar-related circuits such as around the mitral annulus, prior surgical scars, or ablation lesion gaps, often producing ECG patterns distinct from the classic sawtooth morphology.

#165 · Clinical Arrhythmias Atrial

In which patient population is atypical atrial flutter especially common?

Answer

Patients with prior AF ablation, prior atriotomy or cardiac surgery, or significant atrial scarring, since surgical or ablation-related scar provides substrate for non-cavotricuspid isthmus-dependent macro-reentrant circuits.

#166 · Clinical Arrhythmias Atrial

What is the general approach to pharmacologic rate control in atrial flutter?

Answer

Beta-blockers or nondihydropyridine calcium channel blockers are used similarly to AF, though atrial flutter is often relatively resistant to adequate rate control due to its organized, rapid atrial rate and frequent fixed AV conduction ratios.

#167 · Clinical Arrhythmias Atrial

Why is pharmacologic rhythm control generally less effective for typical atrial flutter compared with catheter ablation?

Answer

Antiarrhythmic drugs have modest efficacy in maintaining sinus rhythm for atrial flutter and can paradoxically slow the flutter rate enough to allow 1:1 AV conduction, whereas cavotricuspid isthmus ablation offers a high long-term cure rate.

#168 · Clinical Arrhythmias Atrial

What is the target endpoint of catheter ablation for typical atrial flutter?

Answer

Creation of a linear ablation lesion across the cavotricuspid isthmus that produces bidirectional conduction block, confirmed by differential pacing maneuvers demonstrating both directions of block across the isthmus line.

#169 · Clinical Arrhythmias Atrial

What is the expected long-term success rate of cavotricuspid isthmus ablation for typical atrial flutter?

Answer

Cavotricuspid isthmus ablation achieves acute bidirectional block in the great majority of cases and has a high long-term success rate exceeding 90 percent, making it a preferred first-line therapy over long-term antiarrhythmic drugs.

#170 · Clinical Arrhythmias Atrial

How is ablation of atypical atrial flutter generally approached differently from typical flutter ablation?

Answer

Atypical flutter ablation requires detailed electroanatomic mapping and activation or entrainment mapping to identify the specific reentrant circuit and its critical isthmus, since the circuit location varies by patient scar anatomy rather than being anatomically fixed like the cavotricuspid isthmus.

#171 · Clinical Arrhythmias Atrial

What defines focal atrial tachycardia mechanistically?

Answer

Focal atrial tachycardia arises from a single, discrete atrial site through enhanced automaticity, triggered activity, or micro-reentry, producing a regular atrial rhythm that originates from and spreads centrifugally away from that focus.

#172 · Clinical Arrhythmias Atrial

What ECG feature helps distinguish focal atrial tachycardia from sinus tachycardia?

Answer

Focal atrial tachycardia typically shows a P-wave morphology distinctly different from the patient's baseline sinus P wave, along with an abrupt rather than gradual onset and offset of the tachycardia.

#173 • Clinical Arrhythmias Atrial

What is the warm-up and cool-down pattern sometimes seen in automatic focal atrial tachycardia?

Answer

Automatic focal atrial tachycardia often shows a gradual increase in rate at onset and gradual decrease before termination, reflecting the physiology of an automatic focus rather than the abrupt onset typical of reentrant arrhythmias.

#174 • Clinical Arrhythmias Atrial

What is the first-line pharmacologic approach for symptomatic focal atrial tachycardia?

Answer

Beta-blockers or nondihydropyridine calcium channel blockers are typically first-line for rate control or suppression, with class IC or III antiarrhythmics considered for rhythm control if AV nodal blockers are insufficient.

#175 · Clinical Arrhythmias Atrial

What is the response of an automatic focal atrial tachycardia to adenosine?

Answer

Adenosine often transiently slows or terminates automatic focal atrial tachycardia in some cases by suppressing triggered or automatic activity, though it may only produce transient AV block without terminating the atrial focus itself, unlike its reliable termination of AV nodal-dependent reentrant tachycardias.

#176 · Clinical Arrhythmias Atrial

How does P-wave axis help localize a focal atrial tachycardia to the right versus left atrium?

Answer

A positive P wave in lead V1 suggests a left atrial origin, while a negative or biphasic P wave in V1 suggests a right atrial origin, since V1 reflects the initial direction of atrial activation relative to the interatrial septum.

#177 · Clinical Arrhythmias Atrial

What does a negative P wave in the inferior leads during focal atrial tachycardia suggest about the site of origin?

Answer

A negative P wave in leads II, III, and aVF suggests an inferior atrial focus, such as near the coronary sinus ostium, tricuspid annulus, or crista terminalis, since inferior foci activate the atria in a caudocranial direction.

#178 · Clinical Arrhythmias Atrial

What are common anatomic sites of origin for right atrial focal atrial tachycardia?

Answer

The crista terminalis, tricuspid annulus, coronary sinus ostium, and superior vena cava junction with the right atrium are common right-sided sites given their arrhythmogenic potential from anisotropic fiber orientation and automaticity.

#179 · Clinical Arrhythmias Atrial

What are common anatomic sites of origin for left atrial focal atrial tachycardia?

Answer

The pulmonary vein ostia, left atrial appendage, and mitral annulus are common left-sided sites of focal atrial tachycardia origin.

#180 · Clinical Arrhythmias Atrial

What ablation strategy is used to treat focal atrial tachycardia once the site of origin is identified?

Answer

Activation mapping is used to identify the site of earliest atrial activation relative to P-wave onset, and radiofrequency or cryoablation is delivered at that focal point to eliminate the arrhythmogenic source, often guided by a pace-mapping match to the clinical P-wave morphology.

#181 · Clinical Arrhythmias Supraventricular Tachycardias

What is the difference between WPW pattern and WPW syndrome?

Answer

WPW pattern refers to the ECG findings of pre-excitation (short PR, delta wave, wide QRS) in an asymptomatic patient, while WPW syndrome requires both the ECG pattern and documented episodes of symptomatic tachyarrhythmia.

#182 · Clinical Arrhythmias Supraventricular Tachycardias

What three classic ECG findings define ventricular pre-excitation?

Answer

A short PR interval (less than 120 ms), a delta wave representing slurred upstroke of the QRS, and a widened QRS complex from fusion of conduction over the accessory pathway and the AV node.

#183 · Clinical Arrhythmias Supraventricular Tachycardias

Why do accessory pathways cause a short PR interval?

Answer

The accessory pathway bypasses the AV node's decremental conduction delay, allowing the ventricle to be activated earlier than it would be through the normal His-Purkinje system.

#184 · Clinical Arrhythmias Supraventricular Tachycardias

Which class of antiarrhythmic drugs is contraindicated in pre-excited atrial fibrillation?

Answer

AV nodal blocking agents (adenosine, beta blockers, non-dihydropyridine calcium channel blockers, and digoxin) are contraindicated because blocking the AV node preferentially shifts conduction down the accessory pathway, which can accelerate the ventricular rate and precipitate ventricular fibrillation.

#185 · Clinical Arrhythmias Supraventricular Tachycardias

What is the preferred acute treatment for hemodynamically stable pre-excited atrial fibrillation?

Answer

Intravenous procainamide or ibutilide, which slow conduction through the accessory pathway; if the patient is unstable, immediate synchronized cardioversion is indicated.

#186 · Clinical Arrhythmias Supraventricular Tachycardias

What EP study finding identifies a high-risk accessory pathway in an asymptomatic patient with WPW pattern?

Answer

A shortest pre-excited RR interval during induced atrial fibrillation of less than 250 ms indicates rapid antegrade conduction capacity and identifies a pathway at high risk for sudden cardiac death.

#187 · Clinical Arrhythmias Supraventricular Tachycardias

What is the effective refractory period cutoff of an accessory pathway that suggests low risk for sudden death?

Answer

An antegrade effective refractory period longer than 250 ms is generally considered low risk, since the pathway cannot conduct atrial fibrillation to the ventricle fast enough to cause hemodynamic collapse.

#188 · Clinical Arrhythmias Supraventricular Tachycardias

What non-invasive finding suggests low-risk pre-excitation without EP study?

Answer

Abrupt, intermittent loss of the delta wave during exercise testing or on ambulatory monitoring suggests a long antegrade refractory period and lower risk of rapid conduction during atrial fibrillation.

#189 · Clinical Arrhythmias Supraventricular Tachycardias

In orthodromic AVRT, what is the direction of impulse travel through the accessory pathway?

Answer

In orthodromic AVRT the impulse travels antegrade down the AV node and His-Purkinje system to activate the ventricle, then returns retrograde up the accessory pathway to activate the atrium, producing a narrow QRS complex.

#190 · Clinical Arrhythmias Supraventricular Tachycardias

In antidromic AVRT, what is the direction of impulse travel and resulting QRS morphology?

Answer

The impulse travels antegrade down the accessory pathway to activate the ventricle directly, then returns retrograde up the AV node, producing a wide, fully pre-excited QRS that can mimic ventricular tachycardia.

#191 · Clinical Arrhythmias Supraventricular Tachycardias

Why is antidromic AVRT less common than orthodromic AVRT?

Answer

Antidromic AVRT requires the accessory pathway to conduct antegrade and the AV node to conduct retrograde, a combination that is less common than the reverse; it accounts for only about 5 to 10 percent of AVRT cases.

#192 · Clinical Arrhythmias Supraventricular Tachycardias

How does adenosine typically affect orthodromic AVRT?

Answer

Adenosine transiently blocks the AV node, which is part of the reentrant circuit in orthodromic AVRT, and terminates the tachycardia in most patients.

#193 · Clinical Arrhythmias Supraventricular Tachycardias

What is the target of catheter ablation for a manifest accessory pathway?

Answer

Radiofrequency or cryoablation is delivered at the atrial or ventricular insertion site of the pathway, identified by the earliest local activation and by a sharp accessory pathway potential recorded between the atrial and ventricular electrograms.

#194 · Clinical Arrhythmias Supraventricular Tachycardias

What ablation approach is preferred for a septal accessory pathway near the AV node?

Answer

Cryoablation is often preferred for anteroseptal and mid-septal pathways because of its lower risk of causing complete heart block compared to radiofrequency energy, and because cryomapping (a reversible partial freeze) allows testing before a permanent lesion is delivered.

#195 · Clinical Arrhythmias Supraventricular Tachycardias

What is the approach for ablating a left-sided accessory pathway?

Answer

A left-sided pathway is typically ablated using a retrograde transaortic approach or a transseptal approach from the right atrium into the left atrium, targeting the mitral annulus at the pathway insertion site.

#196 · Clinical Arrhythmias Supraventricular Tachycardias

What is a fasciculoventricular pathway?

Answer

A fasciculoventricular pathway connects the His bundle or a bundle branch directly to the ventricular myocardium, producing a fixed degree of pre-excitation that does not change with atrial pacing and carries no risk of tachyarrhythmia since it cannot form a reentrant circuit with the atrium.

#197 · Clinical Arrhythmias Supraventricular Tachycardias

How is a fasciculoventricular pathway distinguished from a typical atrioventricular accessory pathway on EP study?

Answer

The degree of pre-excitation from a fasciculoventricular pathway remains constant regardless of atrial pacing site or rate, because the pathway originates distal to the AV node rather than from the atrium.

#198 · Clinical Arrhythmias Supraventricular Tachycardias

What is the significance of multiple accessory pathways in one patient?

Answer

Multiple accessory pathways occur in about 10 to 15 percent of WPW patients and are more common in Ebstein anomaly; they increase the complexity of ablation and can support unusual tachycardia circuits using different pathways for antegrade and retrograde conduction.

#199 · Clinical Arrhythmias Supraventricular Tachycardias

What congenital heart disease is classically associated with multiple right-sided accessory pathways?

Answer

Ebstein anomaly, in which the tricuspid valve is displaced apically, is strongly associated with multiple right free wall and posteroseptal accessory pathways.

#200 · Clinical Arrhythmias Supraventricular Tachycardias

What is typical AVNRT?

Answer

Typical (slow-fast) AVNRT uses the slow AV nodal pathway for antegrade conduction and the fast pathway for retrograde conduction, producing a short RP interval with the retrograde P wave often buried in or just after the QRS.

#201 · Clinical Arrhythmias Supraventricular Tachycardias

What is atypical AVNRT?

Answer

Atypical (fast-slow or slow-slow) AVNRT uses the fast or a second slow pathway for antegrade conduction and the slow pathway for retrograde conduction, producing a long RP interval with the retrograde P wave visible well before the next QRS.

#202 · Clinical Arrhythmias Supraventricular Tachycardias

What is dual AV nodal physiology?

Answer

Dual AV nodal physiology describes the presence of two functionally distinct conduction pathways within or near the AV node, a fast pathway with a longer refractory period and a slow pathway with a shorter refractory period, which is the substrate for AVNRT.

#203 · Clinical Arrhythmias Supraventricular Tachycardias

How is dual AV nodal physiology demonstrated during an EP study?

Answer

A jump in the AH interval of 50 ms or more in response to a 10 ms decrement in atrial extrastimulus coupling interval demonstrates a sudden shift from fast to slow pathway conduction.

#204 · Clinical Arrhythmias Supraventricular Tachycardias

Why can AVNRT occur even without a demonstrable AH jump?

Answer

Some patients have dual pathways with overlapping conduction properties, so AVNRT can be induced by a critically timed extrastimulus without a discrete jump, particularly when isoproterenol is used to facilitate induction.

#205 · Clinical Arrhythmias Supraventricular Tachycardias

What is the classic ECG finding of typical AVNRT that helps distinguish it from AVRT?

Answer

A pseudo-R prime in lead V1 or a pseudo-S wave in the inferior leads, caused by the retrograde P wave distorting the terminal portion of the QRS, is characteristic of typical AVNRT.

#206 · Clinical Arrhythmias Supraventricular Tachycardias

What is the ablation target for typical AVNRT?

Answer

The slow pathway is targeted, typically in the posteroseptal region near the coronary sinus os along the tricuspid annulus, identified by a characteristic slow pathway potential and confirmed by junctional ectopy during energy delivery.

#207 · Clinical Arrhythmias Supraventricular Tachycardias

Why is junctional ectopy during slow pathway ablation considered a favorable sign?

Answer

Junctional beats indicate the ablation catheter is affecting slow pathway tissue near the compact AV node, and their appearance with preserved 1:1 retrograde VA conduction confirms intact AV conduction while suggesting effective lesion formation.

#208 · Clinical Arrhythmias Supraventricular Tachycardias

What complication is monitored for continuously during slow pathway ablation?

Answer

Complete heart block, because the slow pathway lies close to the compact AV node; energy delivery is stopped immediately if PR prolongation, AV block, or loss of 1:1 VA conduction during junctional rhythm occurs.

#209 · Clinical Arrhythmias Supraventricular Tachycardias

What is the approximate long-term success rate and recurrence rate of slow pathway ablation for AVNRT?

Answer

Acute success exceeds 95 percent, with a long-term recurrence rate of approximately 2 to 5 percent.

#210 · Clinical Arrhythmias Supraventricular Tachycardias

What is junctional tachycardia?

Answer

Junctional tachycardia is an automatic or triggered arrhythmia arising from the AV junction or proximal His bundle, producing a narrow QRS rhythm typically faster than 100 bpm with AV dissociation or retrograde atrial activation.

#211 · Clinical Arrhythmias Supraventricular Tachycardias

What clinical settings commonly produce junctional tachycardia?

Answer

Digoxin toxicity, cardiac surgery (postoperative junctional ectopic tachycardia), myocardial ischemia, and catheter ablation near the AV node commonly produce junctional tachycardia.

#212 · Clinical Arrhythmias Supraventricular Tachycardias

How does junctional tachycardia typically respond to vagal maneuvers or adenosine?

Answer

Because junctional tachycardia arises from automaticity within the AV junction rather than a reentrant circuit involving the AV node, it usually does not terminate with adenosine or vagal maneuvers, though transient AV block may be seen.

#213 · Clinical Arrhythmias Supraventricular Tachycardias

How is postoperative junctional ectopic tachycardia typically managed in the acute setting?

Answer

Management focuses on correcting electrolyte and temperature abnormalities, minimizing catecholamine support, and using agents like amiodarone or beta blockers to slow the rate, since the rhythm is usually self-limited and ablation is reserved for refractory cases.

#214 · Clinical Arrhythmias Supraventricular Tachycardias

What ECG feature helps distinguish junctional tachycardia from AVNRT?

Answer

Junctional tachycardia often shows a warm-up and cool-down pattern with variable rate typical of automatic rhythms, and AV dissociation with a slower atrial rate than the junctional rate, unlike the abrupt onset and offset of reentrant AVNRT.

#215 · Clinical Arrhythmias Supraventricular Tachycardias

What is the mechanism of focal atrial tachycardia?

Answer

Focal atrial tachycardia arises from a single ectopic focus in the atrium due to abnormal automaticity, triggered activity, or micro-reentry, producing a P wave with a morphology distinct from sinus rhythm that precedes each QRS.

#216 · Clinical Arrhythmias Supraventricular Tachycardias

What is the most common mechanism underlying paroxysmal supraventricular tachycardia overall?

Answer

AVNRT is the most common mechanism, accounting for roughly 50 to 60 percent of paroxysmal SVT cases, followed by AVRT using a concealed or manifest accessory pathway.

#217 · Clinical Arrhythmias Supraventricular Tachycardias

What is a concealed accessory pathway?

Answer

A concealed accessory pathway conducts only retrograde from ventricle to atrium and produces no delta wave on baseline ECG, but it can still support orthodromic AVRT.

#218 · Clinical Arrhythmias Supraventricular Tachycardias

Why does a concealed accessory pathway not cause pre-excitation on the surface ECG?

Answer

Because it lacks antegrade conduction capability, the ventricle is activated only through the normal AV node and His-Purkinje system during sinus rhythm, producing a normal QRS with no delta wave.

#219 · Clinical Arrhythmias Supraventricular Tachycardias

What vagal maneuver is first-line for terminating a stable narrow-complex SVT?

Answer

The modified Valsalva maneuver, which includes leg raise and supine repositioning after straining, has been shown to increase termination rates compared to standard Valsalva alone.

#220 · Clinical Arrhythmias Supraventricular Tachycardias

What is the standard adenosine dosing strategy for terminating SVT?

Answer

An initial rapid intravenous push of 6 mg followed by a saline flush, escalating to 12 mg if the first dose fails to terminate the tachycardia, given through a large proximal vein.

#221 · Clinical Arrhythmias Supraventricular Tachycardias

Why should adenosine be used cautiously in a patient with a wide-complex tachycardia of unclear etiology and known WPW?

Answer

If the wide-complex tachycardia is actually pre-excited atrial fibrillation or antidromic AVRT, adenosine's AV nodal blocking effect could enhance conduction over the accessory pathway and precipitate ventricular fibrillation.

#222 · Clinical Arrhythmias Supraventricular Tachycardias

What is the differential diagnosis for a regular narrow-complex tachycardia with no visible P waves?

Answer

The leading considerations are AVNRT (P wave hidden within the QRS) and orthodromic AVRT with a very short RP interval, since both can produce a retrograde P wave that is not discernible on standard leads.

#223 · Clinical Arrhythmias Supraventricular Tachycardias

What is the differential diagnosis for a long RP tachycardia?

Answer

Atypical AVNRT, orthodromic AVRT using a slowly conducting concealed pathway (permanent junctional reciprocating tachycardia), atrial tachycardia, and sinus tachycardia should all be considered.

#224 · Clinical Arrhythmias Supraventricular Tachycardias

What is permanent junctional reciprocating tachycardia (PJRT)?

Answer

PJRT is an incessant form of orthodromic AVRT using a slowly conducting, decremental, typically posteroseptal concealed accessory pathway, producing a long RP tachycardia that can lead to tachycardia-induced cardiomyopathy if untreated.

#225 · Clinical Arrhythmias Supraventricular Tachycardias

Why is PJRT often mistaken for inappropriate sinus tachycardia or a cardiomyopathy of unknown cause?

Answer

Its incessant, often asymptomatic nature and negative P waves in the inferior leads mimicking a low atrial or coronary sinus rhythm can obscure the diagnosis until echocardiography reveals unexplained left ventricular dysfunction.

#226 · Clinical Arrhythmias Supraventricular Tachycardias

What is the AH interval and what does it represent?

Answer

The AH interval is measured from the earliest atrial deflection to the onset of the His bundle deflection on intracardiac recording and represents conduction time through the AV node.

#227 · Clinical Arrhythmias Supraventricular Tachycardias

What is the HV interval and what does it represent?

Answer

The HV interval is measured from the onset of the His bundle deflection to the earliest ventricular activation and represents conduction time through the His-Purkinje system; a normal value is approximately 35 to 55 ms.

#228 · Clinical Arrhythmias Supraventricular Tachycardias

How does the ventriculoatrial (VA) activation pattern differ between AVNRT and septal AVRT during tachycardia?

Answer

AVNRT typically shows concentric, near-simultaneous atrial activation earliest at the His bundle region, while septal AVRT via an accessory pathway shows eccentric or earlier activation at the pathway's specific atrial insertion site.

#229 · Clinical Arrhythmias Supraventricular Tachycardias

What is the parahisian pacing maneuver used to distinguish?

Answer

Parahisian pacing helps differentiate AVNRT or nodofascicular pathway conduction from a true septal accessory pathway by comparing the VA interval and atrial activation pattern during pacing that captures the His bundle versus pacing that captures only local ventricular myocardium.

#230 · Clinical Arrhythmias Supraventricular Tachycardias

What is the significance of a VA interval greater than 70 ms measured from the onset of the QRS to the earliest atrial activation during SVT?

Answer

A VA interval of 70 ms or more favors AVRT using an accessory pathway, since AVNRT typically has a very short VA interval due to near-simultaneous atrial and ventricular activation through the compact node.

#231 · Clinical Arrhythmias Supraventricular Tachycardias

What is the effect of a bundle branch block during orthodromic AVRT using an ipsilateral accessory pathway?

Answer

A bundle branch block ipsilateral to the accessory pathway lengthens the tachycardia cycle length because the ventricular wavefront must travel a longer path to reach the pathway's ventricular insertion, a finding known as the Coumel sign that supports the diagnosis of AVRT.

#232 · Clinical Arrhythmias Supraventricular Tachycardias

What is atrio-His (AH) block and why is it clinically reassuring during ablation near the AV node?

Answer

Block occurring at the AV node (AH block) rather than below the His bundle (infra-Hisian block) is generally benign and does not indicate a risk of complete heart block, since it reflects normal decremental AV nodal properties rather than damage to the conduction system.

#233 · Clinical Arrhythmias Supraventricular Tachycardias

How is atrial tachycardia distinguished from AVNRT or AVRT using entrainment from the ventricle?

Answer

Ventricular entrainment during atrial tachycardia typically shows a VAAV response pattern with atrial activation dissociated from the pacing, whereas AVNRT and AVRT show a VAV response since the atrium is an obligatory part of the reentrant circuit.

#234 · Clinical Arrhythmias Supraventricular Tachycardias

What is the postpacing interval (PPI) minus tachycardia cycle length (TCL) criterion used for?

Answer

A PPI minus TCL of 115 ms or less after ventricular entrainment of a narrow-complex SVT suggests the ventricle is part of the reentrant circuit, favoring AVNRT or AVRT over atrial tachycardia.

#235 · Clinical Arrhythmias Supraventricular Tachycardias

What defines antidromic AVRT electrophysiologically compared to pre-excited atrial fibrillation?

Answer

Antidromic AVRT is a regular reentrant tachycardia using the accessory pathway antegrade and the AV node (or another pathway) retrograde, while pre-excited atrial fibrillation is an irregular rhythm in which atrial fibrillation conducts antegrade over the accessory pathway.

#236 · Clinical Arrhythmias Supraventricular Tachycardias

Why can antidromic AVRT be mistaken for ventricular tachycardia?

Answer

Because the ventricle is activated entirely through the accessory pathway, the QRS is maximally pre-excited and wide, mimicking the morphology of monomorphic VT, especially in a hemodynamically unstable patient.

#237 · Clinical Arrhythmias Supraventricular Tachycardias

What is the appropriate acute treatment for a hemodynamically unstable wide-complex tachycardia of uncertain mechanism in a WPW patient?

Answer

Immediate synchronized electrical cardioversion, since pharmacologic AV nodal blockade could be harmful if the rhythm is pre-excited atrial fibrillation or antidromic AVRT.

#238 · Clinical Arrhythmias Supraventricular Tachycardias

What is the mechanism of atriofascicular (Mahaim) pathway tachycardia?

Answer

An atriofascicular pathway has decremental, AV node-like conduction properties and inserts near the right bundle branch, supporting an antidromic-type tachycardia with a left bundle branch block morphology during antegrade conduction and typically using the AV node retrograde.

#239 · Clinical Arrhythmias Supraventricular Tachycardias

What ECG morphology is characteristic of Mahaim fiber-mediated tachycardia?

Answer

A left bundle branch block pattern with a relatively normal frontal plane axis is characteristic, reflecting right-sided pre-excitation from the atriofascicular pathway's insertion near the right ventricular apex.

#240 · Clinical Arrhythmias Supraventricular Tachycardias

Why do most patients with WPW pattern never develop symptomatic arrhythmia?

Answer

The presence of an accessory pathway alone does not guarantee formation of a reentrant circuit; symptomatic tachycardia requires a critically timed premature beat encountering differential refractoriness between the AV node and pathway, which does not occur in all patients.

#241 · Clinical Arrhythmias Supraventricular Tachycardias

What is the recommended management for an asymptomatic young patient found to have WPW pattern with high-risk features on non-invasive testing?

Answer

Referral for electrophysiology study with consideration of catheter ablation is recommended given the risk of sudden cardiac death, particularly in patients who wish to participate in competitive athletics.

#242 · Clinical Arrhythmias Supraventricular Tachycardias

What is the recurrence risk after accessory pathway ablation compared to slow pathway ablation for AVNRT?

Answer

Accessory pathway ablation generally has a slightly higher recurrence rate than slow pathway ablation, in the range of 5 to 8 percent depending on pathway location, with septal and epicardial pathways being more prone to recurrence.

#243 · Clinical Arrhythmias Supraventricular Tachycardias

What anatomic structure can harbor an epicardial accessory pathway requiring coronary sinus mapping?

Answer

A left posteroseptal or left free wall pathway may course through the coronary sinus musculature or a coronary sinus diverticulum, requiring mapping and ablation from within the coronary sinus or an adjacent venous branch.

#244 · Clinical Arrhythmias Supraventricular Tachycardias

What complication is specifically associated with ablation near the coronary sinus or its branches?

Answer

Coronary sinus perforation or injury to an adjacent coronary artery, since the coronary sinus and its tributaries run in close proximity to the circumflex artery along the mitral annulus.

#245 · Clinical Arrhythmias Supraventricular Tachycardias

What is the risk of AV block with ablation of an anteroseptal accessory pathway compared to other locations?

Answer

Anteroseptal pathways carry the highest risk of inadvertent AV block among accessory pathway locations because of their proximity to the compact AV node and His bundle, favoring a cryoablation approach.

#246 · Clinical Arrhythmias Supraventricular Tachycardias

What test can help confirm loss of accessory pathway conduction immediately after ablation?

Answer

Programmed atrial and ventricular stimulation, along with observation for at least 20 to 30 minutes, is used to confirm loss of both antegrade and retrograde pathway conduction and absence of inducible tachycardia.

#247 · Clinical Arrhythmias Supraventricular Tachycardias

What is the mechanism by which isoproterenol facilitates induction of SVT during EP study?

Answer

Isoproterenol enhances sympathetic tone, shortening refractory periods and enhancing automaticity and triggered activity, which can unmask dual AV nodal physiology or facilitate induction of AVNRT, AVRT, or atrial tachycardia when baseline attempts fail.

#248 · Clinical Arrhythmias Supraventricular Tachycardias

What is the significance of an atrial echo beat during programmed stimulation?

Answer

An atrial echo beat, in which a single atrial extrastimulus returns to activate the atrium after conducting down the slow pathway and back up the fast pathway, confirms the substrate for AVNRT even if sustained tachycardia is not induced.

#249 · Clinical Arrhythmias Supraventricular Tachycardias

What distinguishes slow-slow atypical AVNRT from typical AVNRT electrophysiologically?

Answer

Slow-slow AVNRT uses a second, distinct slow pathway for retrograde conduction rather than the fast pathway, often with atrial activation earliest near the coronary sinus os rather than the His bundle region.

#250 · Clinical Arrhythmias Supraventricular Tachycardias

How does carotid sinus massage terminate reentrant SVT?

Answer

Carotid sinus massage increases vagal tone, which slows or blocks conduction through the AV node, interrupting the reentrant circuit in AVNRT and AV node-dependent AVRT.

#251 · Clinical Arrhythmias Supraventricular Tachycardias

What is the role of ibutilide in pre-excited atrial fibrillation?

Answer

Ibutilide prolongs the refractory period of the accessory pathway and can chemically cardiovert atrial fibrillation to sinus rhythm, but it carries a risk of QT prolongation and torsades de pointes requiring continuous monitoring.

#252 · Clinical Arrhythmias Supraventricular Tachycardias

What is the long-term risk in a WPW patient who is never treated with ablation or medication?

Answer

The lifetime risk of sudden cardiac death from rapid conduction of atrial fibrillation over a high-risk accessory pathway is estimated at approximately 0.1 to 0.6 percent per year, which is why risk stratification is important even in asymptomatic patients.

#253 · Clinical Arrhythmias Supraventricular Tachycardias

What ECG clue suggests a left free wall accessory pathway location?

Answer

A negative delta wave in lead I with a positive delta wave in the inferior leads and a dominant R wave in V1 suggests a left free wall pathway, since ventricular pre-excitation begins on the left side and vectors rightward.

#254 · Clinical Arrhythmias Supraventricular Tachycardias

What ECG clue suggests a right free wall accessory pathway location?

Answer

A negative or biphasic delta wave in the right precordial leads with a dominant S wave pattern suggests a right free wall pathway, since the initial ventricular activation begins on the right side.

#255 · Clinical Arrhythmias Supraventricular Tachycardias

Why do algorithms localizing accessory pathways from the surface ECG become less reliable with multiple pathways or minimal pre-excitation?

Answer

These algorithms depend on a clean, fully expressed delta wave vector from a single pathway; fusion between fast AV nodal and pathway conduction, or the presence of a second pathway, distorts the vector and reduces localization accuracy.

#256 · Clinical Arrhythmias Supraventricular Tachycardias

What is the appropriate management strategy for a symptomatic patient with documented AVNRT who declines ablation?

Answer

Pharmacologic therapy with an AV nodal blocking agent such as a beta blocker or non-dihydropyridine calcium channel blocker can be used for rate control and suppression, though ablation offers a definitive cure with high success and low recurrence.

#257 · Clinical Arrhythmias Supraventricular Tachycardias

What is the differential consideration when a patient has inducible dual AV nodal physiology but no sustained AVNRT during EP study despite clinical symptoms of palpitations?

Answer

The operator should consider that the clinical arrhythmia may be a different mechanism, such as atrial tachycardia or an unmapped accessory pathway, and continue to search for alternative substrates rather than assuming AVNRT explains the symptoms.

#258 · Clinical Arrhythmias Supraventricular Tachycardias

What is the mechanism of the 'jump-shift' phenomenon seen during atrial extrastimulus testing in dual AV nodal physiology?

Answer

As the coupling interval of the atrial extrastimulus shortens, the fast pathway becomes refractory first, causing conduction to abruptly shift to the slower but less refractory slow pathway, producing a sudden increase in the AH interval.

#259 · Clinical Arrhythmias Supraventricular Tachycardias

What is the typical cycle length range for AVNRT compared to atrial flutter?

Answer

AVNRT typically has a cycle length between 300 and 400 ms (rate 150 to 200 bpm), while typical atrial flutter has an atrial cycle length around 200 ms with variable AV conduction ratios affecting the ventricular rate.

#260 · Clinical Arrhythmias Supraventricular Tachycardias

What finding during entrainment from the atrium helps distinguish AVRT from AVNRT when an accessory pathway is suspected but not obviously visible?

Answer

A change in the VA interval or tachycardia cycle length with development of bundle branch block, or a stimulus-to-QRS to VA interval difference exceeding 85 ms compared to tachycardia, supports the presence of an accessory pathway rather than AVNRT.

#261 · Clinical Arrhythmias Supraventricular Tachycardias

Why might catheter ablation fail to eliminate an accessory pathway on the first attempt?

Answer

Common reasons include an epicardial pathway location inaccessible from the endocardium, inadequate catheter contact or stability, an anatomically difficult location such as within the coronary sinus, or transient pathway suppression (edema) mistaken for permanent block.

#262 · Clinical Arrhythmias Supraventricular Tachycardias

What is the recommended waiting period after apparent successful accessory pathway ablation before final confirmation of block?

Answer

A waiting period of at least 20 to 30 minutes with repeat stimulation is recommended to exclude transient conduction block from tissue edema masquerading as a permanent lesion.

#263 · Clinical Arrhythmias Supraventricular Tachycardias

What is the clinical significance of a decremental accessory pathway?

Answer

A decremental (Mahaim-type or atriofascicular) pathway conducts more slowly as the pacing rate increases, similar to AV nodal tissue, distinguishing it from typical fast, non-decremental Kent bundle pathways and altering the expected tachycardia mechanism and ECG appearance.

#264 · Clinical Arrhythmias Supraventricular Tachycardias

What is the general recommendation for catheter ablation versus long-term drug therapy for symptomatic AVRT or AVNRT in an otherwise healthy adult?

Answer

Catheter ablation is generally preferred as first-line therapy over long-term antiarrhythmic drug therapy because of its high cure rate, low complication rate, and avoidance of lifelong medication.

#265 · Clinical Arrhythmias Supraventricular Tachycardias

Why is genetic testing considered in a young patient presenting with WPW pattern and evidence of cardiac hypertrophy?

Answer

WPW associated with left ventricular hypertrophy in a young patient can indicate PRKAG2 cardiac syndrome, a glycogen storage cardiomyopathy that mimics hypertrophic cardiomyopathy and carries its own risk of progressive conduction disease, so genetic evaluation is warranted.

#266 · Clinical Arrhythmias Ventricular

On a 12-lead ECG, what QRS pattern in the precordial leads suggests a PVC or VT origin from the right ventricle?

Answer

A left bundle branch block pattern (dominant S wave in V1) suggests a right ventricular origin, since activation must cross the septum to reach the left ventricle.

#267 · Clinical Arrhythmias Ventricular

What QRS morphology in V1 suggests a left ventricular origin of a PVC or VT?

Answer

A right bundle branch block pattern, typically a dominant R wave or qR/Rsr' complex in V1, suggests the impulse originates in the left ventricle and activates the right ventricle late.

#268 · Clinical Arrhythmias Ventricular

How do you use the inferior lead axis (II, III, aVF) to localize a PVC's superior versus inferior origin?

Answer

A positive (upright) axis in the inferior leads points to a superior origin (near the outflow tracts), with activation traveling downward toward the inferior leads, while a negative axis in the inferior leads points to an inferior origin, with activation traveling upward away from those leads.

#269 · Clinical Arrhythmias Ventricular

What precordial transition pattern helps distinguish right ventricular outflow tract (RVOT) from left ventricular outflow tract (LVOT) PVCs?

Answer

RVOT PVCs typically transition to a dominant R wave at or after V3, while LVOT PVCs show earlier precordial transition, at or before V2.

#270 · Clinical Arrhythmias Ventricular

What is the transition ratio and how is it used to localize outflow tract PVCs?

Answer

The transition ratio compares the R/S ratio in V2 during the PVC to that during sinus rhythm; a ratio greater than 1.5 suggests a left-sided (aortic cusp) origin rather than the RVOT.

#271 · Clinical Arrhythmias Ventricular

What ECG feature suggests a PVC originates from the left posterior fascicle?

Answer

A right bundle branch block pattern with a superior (left/left-superior) axis, since the posterior fascicle exit produces activation that travels anteriorly and superiorly.

#272 · Clinical Arrhythmias Ventricular

What ECG feature suggests a PVC originates from the left anterior fascicle?

Answer

A right bundle branch block pattern with a rightward and inferior axis, reflecting the anterior fascicle's exit site anterosuperior in the septum.

#273 · Clinical Arrhythmias Ventricular

What is the pace-map matching technique for localizing PVC or VT origin?

Answer

A catheter paces from a candidate site and the resulting 12-lead QRS morphology is compared to the clinical PVC or VT; a high percentage match (commonly greater than 95%) suggests proximity to the true origin.

#274 · Clinical Arrhythmias Ventricular

Why can pace-mapping be misleading near the outflow tracts?

Answer

The RVOT and LVOT are anatomically close with overlapping muscular continuity, so similar pace-map morphologies can be produced from sites on either side of the valve, requiring activation mapping for precision.

#275 · Clinical Arrhythmias Ventricular

What activation mapping finding confirms the earliest site of origin for a focal PVC or VT?

Answer

The site with the earliest local bipolar electrogram activation relative to the QRS onset, ideally 20 to 40 milliseconds or more before the surface QRS, confirms proximity to the origin.

#276 · Clinical Arrhythmias Ventricular

What is the class I indication for an invasive EP study in a patient with unexplained syncope and structural heart disease?

Answer

An EP study is indicated in patients with syncope of unexplained cause and structural heart disease or coronary disease when noninvasive testing has not yielded a diagnosis, to assess for inducible ventricular arrhythmia or conduction disease.

#277 · Clinical Arrhythmias Ventricular

What is programmed ventricular stimulation used to assess in ischemic cardiomyopathy?

Answer

It assesses inducibility of sustained monomorphic ventricular tachycardia, which helps risk-stratify patients for sudden cardiac death and can guide decisions about ICD implantation or catheter ablation.

#278 · Clinical Arrhythmias Ventricular

What number of extrastimuli and how many drive-train cycle lengths are typically used in a standard programmed ventricular stimulation protocol?

Answer

A standard protocol uses up to three extrastimuli delivered after drive trains at two or more cycle lengths (commonly 600 and 400 milliseconds), from at least one right ventricular site.

#279 · Clinical Arrhythmias Ventricular

Why is induction of polymorphic VT or ventricular fibrillation with aggressive stimulation protocols considered a nonspecific finding?

Answer

Polymorphic VT or VF can be induced in many patients without clinical ventricular arrhythmia when very short coupling intervals or triple extrastimuli are used, so it lacks specificity for predicting future clinical events.

#280 · Clinical Arrhythmias Ventricular

What is entrainment mapping used to confirm during VT ablation?

Answer

Entrainment mapping confirms that a site is part of the reentrant circuit's protected isthmus by pacing faster than the tachycardia and demonstrating fusion with resumption of the tachycardia at the paced cycle length upon cessation.

#281 · Clinical Arrhythmias Ventricular

What defines concealed entrainment during VT mapping?

Answer

Concealed entrainment occurs when the paced QRS morphology during entrainment is identical to the tachycardia QRS, indicating the pacing site is within the reentrant circuit rather than in bystander tissue.

#282 · Clinical Arrhythmias Ventricular

What is the postpacing interval and how is it used in entrainment mapping?

Answer

The postpacing interval is the time from the last paced stimulus to the next local electrogram after entrainment; when it closely matches the tachycardia cycle length (within about 30 milliseconds), the site is likely within the reentry circuit isthmus.

#283 · Clinical Arrhythmias Ventricular

What does a stimulus-to-QRS interval that closely matches the electrogram-to-QRS interval during sinus rhythm suggest at a VT ablation site?

Answer

It suggests the site lies within a slow-conducting isthmus of the reentry circuit, since the conduction delay from pacing matches the delay observed during the tachycardia.

#284 · Clinical Arrhythmias Ventricular

What are the two main ablation approaches used to eliminate scar-related VT in ischemic cardiomyopathy?

Answer

Endocardial catheter ablation targeting the isthmus within or bordering the infarct scar, and epicardial ablation via a subxiphoid pericardial approach when the arrhythmogenic substrate lies subepicardially or endocardial ablation fails.

#285 · Clinical Arrhythmias Ventricular

What substrate-based ablation strategy is used when VT is not hemodynamically tolerated for activation mapping?

Answer

Substrate mapping identifies areas of low-voltage scar, late potentials, and channels of surviving myocardium during sinus rhythm, and ablation targets these regions or applies scar homogenization without needing to induce the clinical VT.

#286 · Clinical Arrhythmias Ventricular

What voltage thresholds define dense scar versus border zone on a bipolar endocardial voltage map?

Answer

Bipolar electrograms below 0.5 millivolts define dense scar, while electrograms between 0.5 and 1.5 millivolts define the border zone of diseased but partially viable tissue.

#287 · Clinical Arrhythmias Ventricular

Why is inferior wall myocardial infarction more likely to require epicardial ablation than anterior infarction?

Answer

Inferior infarcts more commonly produce a subepicardial or midmyocardial component of scar because of the typical transmural extent and basal involvement, making the arrhythmogenic substrate less accessible from the endocardium alone.

#288 · Clinical Arrhythmias Ventricular

What is a key procedural risk of subxiphoid epicardial access for VT ablation?

Answer

Right ventricular perforation or injury to the coronary arteries or phrenic nerve during needle puncture and sheath advancement into the pericardial space, along with the risk of hemopericardium.

#289 · Clinical Arrhythmias Ventricular

How is the phrenic nerve protected during epicardial VT ablation near its course?

Answer

High-output pacing from the ablation catheter is used to confirm phrenic capture before ablation, and techniques such as balloon or catheter displacement, or air/saline instillation into the pericardial space, can separate the nerve from the ablation site.

#290 · Clinical Arrhythmias Ventricular

What is the role of a left ventricular assist device (LVAD) in facilitating VT ablation?

Answer

An LVAD can provide hemodynamic support that allows prolonged mapping and ablation of poorly tolerated VT by maintaining systemic perfusion despite induced arrhythmia.

#291 · Clinical Arrhythmias Ventricular

Why can VT be more difficult to ablate in a patient with an existing durable LVAD?

Answer

The LVAD cannula and surrounding fibrous tissue can be an arrhythmogenic substrate, cannula suction artifacts can distort electrograms, and altered chamber geometry from unloading can shift arrhythmia circuits and complicate mapping.

#292 · Clinical Arrhythmias Ventricular

What forms of temporary hemodynamic support can be used during high-risk VT ablation for unstable VT?

Answer

Percutaneous options include an intra-aortic balloon pump, a percutaneous left ventricular assist device such as an Impella, and venoarterial extracorporeal membrane oxygenation (ECMO), chosen based on the degree of expected hemodynamic compromise.

#293 · Clinical Arrhythmias Ventricular

How does hemodynamic support during VT ablation change the mapping approach used?

Answer

It permits activation and entrainment mapping to be performed during the induced clinical VT itself rather than relying solely on substrate mapping in sinus rhythm, potentially improving procedural success.

#294 · Clinical Arrhythmias Ventricular

How does the arrhythmogenic substrate in nonischemic dilated cardiomyopathy typically differ in location from ischemic VT substrate?

Answer

Nonischemic cardiomyopathy scar is more often located in the mid-myocardium or epicardium, particularly the basal perivalvular and septal regions, rather than the endocardial infarct-related scar typical of ischemic disease.

#295 · Clinical Arrhythmias Ventricular

Why does VT ablation in nonischemic cardiomyopathy more often require an epicardial or combined endo-epicardial approach?

Answer

Because the fibrotic substrate frequently involves the mid-myocardial or epicardial layers rather than being confined to the endocardium, endocardial-only ablation is less likely to reach the critical circuit components.

#296 · Clinical Arrhythmias Ventricular

What is the significance of perivalvular scar in nonischemic cardiomyopathy VT ablation?

Answer

Perivalvular scar around the mitral or aortic annulus is a common substrate location, and ablation near these regions carries added risk to the coronary arteries, conduction system, and valve function.

#297 · Clinical Arrhythmias Ventricular

How does sarcoidosis-related cardiomyopathy typically present as a substrate for VT ablation?

Answer

Sarcoidosis produces patchy, often basal septal and epicardial granulomatous scar that can be more extensive and multifocal than typical dilated cardiomyopathy scar, frequently requiring combined endo-epicardial mapping.

#298 · Clinical Arrhythmias Ventricular

What imaging modality is particularly useful for characterizing the arrhythmogenic substrate before VT ablation in nonischemic cardiomyopathy?

Answer

Cardiac MRI with late gadolinium enhancement identifies the location and transmural extent of fibrosis, helping predict whether an epicardial approach will be needed and guiding image-integrated mapping.

#299 · Clinical Arrhythmias Ventricular

What is the most common site of origin for idiopathic outflow tract PVCs and VT?

Answer

The right ventricular outflow tract, particularly its septal aspect just below the pulmonic valve, is the most common site in patients without structural heart disease.

#300 · Clinical Arrhythmias Ventricular

What is the typical ECG morphology of idiopathic RVOT VT or PVCs?

Answer

A left bundle branch block pattern with an inferior axis (positive in leads II, III, and aVF), reflecting an origin high and anterior in the right ventricle near the pulmonic valve.

#301 · Clinical Arrhythmias Ventricular

What is the presumed mechanism of idiopathic outflow tract VT?

Answer

Cyclic AMP-mediated triggered activity from delayed afterdepolarizations, which is why it is typically catecholamine-sensitive, exercise- or stress-provoked, and often suppressed by adenosine or beta blockade.

#302 · Clinical Arrhythmias Ventricular

What structures make up the aortic cusps as a source of left-sided outflow tract VT and PVCs?

Answer

The left, right, and noncoronary aortic cusps, with the left and right cusps and the junction between them being the most common sites due to their proximity to ventricular myocardial extensions above the aortic valve.

#303 · Clinical Arrhythmias Ventricular

What is idiopathic fascicular VT and what is its typical mechanism?

Answer

Fascicular VT arises from the Purkinje fiber network, most commonly the left posterior fascicle, and is typically due to a reentrant mechanism involving a verapamil-sensitive slow zone within the fascicular-Purkinje tissue.

#304 · Clinical Arrhythmias Ventricular

What is the classic ECG pattern of left posterior fascicular VT?

Answer

A right bundle branch block morphology with a superior (left axis) frontal plane axis and a relatively narrow QRS duration compared to other forms of VT.

#305 · Clinical Arrhythmias Ventricular

Why is intravenous verapamil useful both diagnostically and therapeutically in fascicular VT?

Answer

Because the reentrant circuit depends on a calcium-dependent slow-conducting zone within the fascicular tissue, verapamil often terminates and suppresses the arrhythmia, supporting the diagnosis and providing acute treatment.

#306 · Clinical Arrhythmias Ventricular

What is the ablation target for fascicular VT identified by a Purkinje potential?

Answer

The site of the earliest presystolic Purkinje potential during tachycardia, or the diastolic potential in the slow zone of the reentrant circuit, is targeted for ablation, often near the mid to apical left posterior fascicle.

#307 · Clinical Arrhythmias Ventricular

What is the typical prognosis of idiopathic outflow tract and fascicular VT in the absence of structural heart disease?

Answer

These arrhythmias generally carry a benign prognosis with low risk of sudden death, though frequent PVCs can cause PVC-induced cardiomyopathy if the burden is high.

#308 · Clinical Arrhythmias Ventricular

What PVC burden threshold is commonly cited as a risk factor for developing PVC-induced cardiomyopathy?

Answer

A PVC burden exceeding approximately 10 to 20% of total heartbeats on ambulatory monitoring is associated with increased risk of left ventricular dysfunction, though other factors such as QRS duration and epicardial origin also contribute.

#309 · Clinical Arrhythmias Ventricular

What defines bradycardia-dependent (pause-dependent) polymorphic VT?

Answer

Polymorphic VT that is initiated by a pause or bradycardia, classically seen in acquired long QT syndrome, where a short-long-short cycle sequence precedes the onset of torsades de pointes.

#310 · Clinical Arrhythmias Ventricular

What is the classic ECG appearance of torsades de pointes?

Answer

A polymorphic ventricular tachycardia in which the QRS complexes appear to twist around the isoelectric baseline, occurring in the setting of a prolonged QT interval.

#311 · Clinical Arrhythmias Ventricular

What is the first-line acute treatment for torsades de pointes?

Answer

Intravenous magnesium sulfate is given regardless of serum magnesium level, along with correction of any electrolyte abnormalities and discontinuation of the offending QT-prolonging drug.

#312 · Clinical Arrhythmias Ventricular

How does temporary pacing help prevent recurrent drug-induced torsades de pointes?

Answer

Atrial or ventricular overdrive pacing at a faster rate shortens the QT interval and eliminates the pause that triggers torsades, serving as a bridge while the causative drug is cleared.

#313 • Clinical Arrhythmias Ventricular

What class of antiarrhythmic drugs is most classically associated with drug-induced torsades de pointes?

Answer

Class IA (such as quinidine and procainamide) and class III (such as sotalol and dofetilide) antiarrhythmic drugs, due to their IKr-blocking effect that prolongs repolarization.

#314 • Clinical Arrhythmias Ventricular

What distinguishes ischemic polymorphic VT and VF from the classic pause-dependent form?

Answer

Ischemic polymorphic VT and VF typically occur without QT prolongation, are triggered by acute myocardial ischemia rather than a pause, and are treated with revascularization and antiischemic therapy rather than pacing or magnesium alone.

#315 · Clinical Arrhythmias Ventricular

What is the mainstay of acute management for polymorphic VT or VF due to active myocardial ischemia?

Answer

Emergent coronary revascularization along with beta blockade and antiischemic medical therapy, since correcting the ischemic trigger is central to preventing recurrence.

#316 · Clinical Arrhythmias Ventricular

What defines electrical storm?

Answer

Three or more separate episodes of sustained VT, VF, or appropriate ICD shocks occurring within a 24-hour period.

#317 · Clinical Arrhythmias Ventricular

What is the initial pharmacologic approach to acute management of electrical storm?

Answer

Intravenous beta blockade (often combined with amiodarone) and sedation to reduce sympathetic drive, since heightened adrenergic tone is a major contributor to recurrent arrhythmia in electrical storm.

#318 · Clinical Arrhythmias Ventricular

What is trigger PVC ablation and when is it used in electrical storm?

Answer

It is catheter ablation targeting the specific PVC morphology that initiates recurrent VT or VF episodes, used when a consistent triggering PVC can be identified and pharmacologic therapy fails to control the storm.

#319 · Clinical Arrhythmias Ventricular

What autonomic modulation procedure can be considered for refractory electrical storm when ablation and drug therapy fail?

Answer

Stellate ganglion blockade or surgical cardiac sympathetic denervation can reduce sympathetic tone and suppress recurrent ventricular arrhythmia in refractory electrical storm.

#320 · Clinical Arrhythmias Ventricular

What is pulsed field ablation (PFA) and how does its mechanism differ from thermal ablation?

Answer

Pulsed field ablation uses high-voltage electrical pulses to induce irreversible electroporation of cell membranes, causing cell death through a nonthermal mechanism rather than the heating (radiofrequency) or freezing (cryoablation) used by conventional energy sources.

#321 · Clinical Arrhythmias Ventricular

What is a proposed advantage of pulsed field ablation for ventricular substrate ablation near critical structures?

Answer

Tissue selectivity that may spare structures such as the phrenic nerve, esophagus, and coronary arteries, since these tissues have differing susceptibility thresholds to electroporation compared to myocardium.

#322 · Clinical Arrhythmias Ventricular

What limitation currently exists for pulsed field ablation in scar-related VT ablation?

Answer

Most pulsed field ablation catheter designs and clinical experience have focused on atrial fibrillation pulmonary vein isolation, and lesion depth in dense ventricular scar or thick myocardium remains an area of ongoing investigation.

#323 · Clinical Arrhythmias Ventricular

How is artificial intelligence being applied to VT substrate mapping?

Answer

Machine learning algorithms are being developed to automatically identify abnormal electrograms, late potentials, and conducting channels within scar from voltage and activation maps, aiming to standardize substrate characterization and speed up procedures.

#324 · Clinical Arrhythmias Ventricular

What is a potential benefit of AI-assisted mapping for VT ablation procedures?

Answer

It may reduce interobserver variability in identifying critical isthmus sites and could shorten procedure and fluoroscopy time by automating substrate analysis that is otherwise labor-intensive and operator-dependent.

#325 · Clinical Arrhythmias Ventricular

What is cardiac (stereotactic) radioablation for VT?

Answer

A noninvasive technique that delivers a single high dose of focused external beam radiation to a mapped arrhythmogenic scar target, using cardiac and respiratory motion tracking, to create an ablative lesion without catheter access.

#326 · Clinical Arrhythmias Ventricular

In whom is cardiac radioablation for VT most often considered?

Answer

Patients with recurrent, refractory scar-related VT who have failed or are poor candidates for repeat catheter ablation, often due to prior failed procedures or prohibitive procedural risk.

#327 · Clinical Arrhythmias Ventricular

What is a key characteristic of the clinical effect of cardiac radioablation compared to catheter ablation?

Answer

The antiarrhythmic effect is delayed, typically developing over weeks to months as the irradiated tissue undergoes fibrotic remodeling, unlike the immediate lesion effect of catheter-based energy sources.

#328 · Clinical Arrhythmias Ventricular

What imaging and planning steps precede cardiac radioablation for VT?

Answer

Prior electroanatomic mapping data (often from a failed catheter procedure) and cardiac imaging are co-registered to define the target volume, and respiratory and cardiac motion are accounted for in treatment planning to protect adjacent structures.

#329 · Clinical Arrhythmias Ventricular

What is the significance of a discrete diastolic potential recorded during VT mapping?

Answer

A discrete, isolated diastolic potential recorded during the diastolic interval of the tachycardia cycle suggests the catheter is positioned within a slow-conducting isthmus of the reentry circuit, a favorable ablation target.

#330 · Clinical Arrhythmias Ventricular

What does a mismatch between the paced QRS morphology and tachycardia QRS morphology during entrainment indicate?

Answer

It indicates the pacing site is not within the reentrant circuit isthmus but rather in a bystander site or an area only loosely connected to the circuit.

#331 · Clinical Arrhythmias Ventricular

What is scar dechanneling in the context of VT substrate ablation?

Answer

An ablation strategy that targets conducting channels of surviving myocardium within dense scar, identified by fractionated or late electrograms, aiming to interrupt potential reentrant pathways even without inducing the clinical VT.

#332 · Clinical Arrhythmias Ventricular

What is scar homogenization as a VT ablation strategy?

Answer

An extensive ablation approach that targets all abnormal (low-voltage and late-potential) electrograms throughout the entire scar region rather than only the presumed isthmus, aiming to eliminate all potential arrhythmogenic substrate.

#333 · Clinical Arrhythmias Ventricular

What is the significance of a late potential recorded on a bipolar electrogram in scar tissue?

Answer

A late potential is a low-amplitude, fractionated signal occurring after the end of the surface QRS complex, reflecting slow conduction through surviving myocardial fibers within scar and marking a potential arrhythmogenic substrate site.

#334 · Clinical Arrhythmias Ventricular

How is unipolar voltage mapping used differently from bipolar voltage mapping in VT substrate assessment?

Answer

Unipolar voltage mapping, with a lower abnormal threshold (commonly below 8.3 millivolts on the endocardium), can detect epicardial or mid-myocardial scar that appears normal on bipolar endocardial mapping, helping predict the need for an epicardial approach.

#335 · Clinical Arrhythmias Ventricular

What is the role of coronary angiography before epicardial VT ablation?

Answer

Coronary angiography is performed to define the course of the epicardial coronary arteries relative to the planned ablation sites, since ablation lesions delivered too close to a coronary artery risk acute occlusion or delayed stenosis.

#336 · Clinical Arrhythmias Ventricular

What safety measure is used when an epicardial ablation site lies close to a coronary artery?

Answer

Ablation is avoided or performed at reduced power within a defined safe distance (commonly within about 5 millimeters) of a coronary artery, sometimes with balloon protection or repositioning of the target.

#337 · Clinical Arrhythmias Ventricular

What is a purse-string or dry pericardial access technique used for in epicardial VT ablation?

Answer

It refers to draining accumulated pericardial fluid or blood at the end of the epicardial procedure and, in some cases, instilling steroid to reduce postprocedural pericarditis, reducing the risk of delayed tamponade.

#338 · Clinical Arrhythmias Ventricular

What is the clinical significance of inducing the clinical VT (matching the patient's documented tachycardia) versus a nonclinical VT during programmed stimulation at ablation?

Answer

Inducing and successfully ablating the clinical VT with confirmed noninducibility afterward is associated with better long-term outcomes, while only eliminating nonclinical VTs leaves the original arrhythmia substrate potentially untreated.

#339 · Clinical Arrhythmias Ventricular

What is the significance of noninducibility of any monomorphic VT at the end of an ablation procedure?

Answer

Complete noninducibility of all previously inducible monomorphic VTs, including nonclinical ones, is associated with the lowest rates of VT recurrence compared to partial success or persistent inducibility.

#340 · Clinical Arrhythmias Ventricular

What baseline factor most strongly predicts the need for combined endocardial-epicardial ablation in nonischemic cardiomyopathy?

Answer

Cardiac MRI evidence of mid-myocardial or epicardial late gadolinium enhancement, particularly in a septal or perivalvular distribution, predicts a higher likelihood that endocardial-only ablation will be insufficient.

#341 • Clinical Arrhythmias Ventricular

What is the mechanism underlying most sustained monomorphic VT in patients with prior myocardial infarction?

Answer

Scar-related reentry, in which surviving myocardial fiber bundles traverse regions of dense fibrosis, creating zones of slow conduction that support a stable reentrant circuit.

#342 • Clinical Arrhythmias Ventricular

How does the border zone channel concept guide ablation lesion placement in ischemic VT?

Answer

Ablation lesions are placed to transect identified conducting channels of viable tissue within or bordering the scar, aiming to interrupt the critical isthmus that supports one or more reentrant VT circuits.

#343 • Clinical Arrhythmias Ventricular

What is the role of a wearable or implantable defibrillator while awaiting definitive VT ablation therapy?

Answer

It provides protection against sudden cardiac death from recurrent ventricular arrhythmia during the period before or after ablation, particularly when ablation success is uncertain or the underlying substrate remains unresolved.

#344 • Clinical Arrhythmias Ventricular

What is the papillary muscle as a source of PVC or VT, and how does it typically appear on ECG?

Answer

Papillary muscle VT and PVCs arise from Purkinje-myocardial junctions within the papillary muscles (most often the left ventricular posteromedial papillary muscle) and typically show a right bundle branch block pattern with a variable axis and often a notched or slurred QRS.

#345 · Clinical Arrhythmias Ventricular

Why can papillary muscle VT and PVCs be more challenging to ablate than fascicular VT?

Answer

The mobile, three-dimensional structure of the papillary muscle makes stable catheter contact difficult, and the arrhythmogenic focus can be located deep within the muscle, sometimes requiring intracardiac echocardiography for guidance and higher power or longer ablation times.

#346 · Clinical Arrhythmias Ventricular

What ECG feature distinguishes idiopathic left ventricular summit VT and PVCs, and why are they challenging to ablate?

Answer

They typically show a left bundle branch block pattern with an inferior axis and often a pattern break or notch in lead I; the summit region sits epicardially near the coronary venous system and major coronary vessels, making safe ablation access difficult.

#347 · Clinical Arrhythmias Ventricular

What defines ventricular fibrillation on the surface ECG?

Answer

A chaotic, disorganized rhythm with no discernible QRS complexes, of varying amplitude and morphology, representing loss of coordinated ventricular electrical and mechanical activity.

#348 · Clinical Arrhythmias Ventricular

What is the significance of a short-coupled PVC that initiates polymorphic VT or VF in an otherwise structurally normal heart?

Answer

It suggests idiopathic short-coupled variant of torsades de pointes or Purkinje-triggered VF, in which a very early-coupled PVC (often from the Purkinje system) initiates the arrhythmia, and the trigger focus can be targeted with catheter ablation.

#349 · Clinical Arrhythmias Ventricular

What is the role of quinidine in idiopathic short-coupled ventricular fibrillation and Brugada-related VF?

Answer

Quinidine can suppress recurrent VF in these conditions by blocking the transient outward potassium current (I_{to}), reducing the substrate for phase 2 reentry and early triggered activity.

#350 · Clinical Arrhythmias Ventricular

What is the general acute hemodynamic support strategy chosen when VT ablation is anticipated to require prolonged mapping of an unstable rhythm without an existing durable LVAD?

Answer

Temporary percutaneous mechanical circulatory support, most often a percutaneous left ventricular assist device or venoarterial ECMO, is placed prophylactically before the procedure to allow safe, extended mapping and ablation of the unstable VT.

#351 · Devices

What is the mechanism by which electromagnetic interference (EMI) can cause a pacemaker to inappropriately inhibit pacing output?

Answer

EMI signals sensed on the lead can be misinterpreted as intrinsic cardiac activity, causing the device to withhold a scheduled pacing pulse, which is dangerous in a pacemaker-dependent patient.

#352 · Devices

How does EMI typically affect an ICD rather than a pacemaker?

Answer

EMI is more likely to be misclassified as ventricular fibrillation, triggering inappropriate shocks, since ICDs use high-frequency noise detection algorithms tuned to sense rapid signals.

#353 · Devices

What is the recommended management for a pacemaker-dependent patient undergoing surgery with monopolar electrocautery near the device?

Answer

Program the device to an asynchronous pacing mode (such as VOO or DOO) or apply a magnet, and have cautery used in short bursts away from the device and leads.

#354 · Devices

Why is bipolar electrocautery generally safer than monopolar for patients with cardiac implantable electronic devices (CIEDs)?

Answer

Bipolar cautery confines current flow between two closely spaced electrode tips at the surgical site, producing much less stray electrical current that can reach the device or leads.

#355 · Devices

What effect does a magnet typically have on a modern ICD?

Answer

It suspends tachyarrhythmia detection and therapy delivery without affecting bradycardia pacing settings, and this effect reverses when the magnet is removed.

#356 · Devices

What effect does a magnet typically have on a modern pacemaker?

Answer

It converts the device to asynchronous pacing at a fixed magnet rate, which can also serve as a quick battery status check depending on the manufacturer.

#357 · Devices

Is MRI safe in patients with a CIED labeled MR conditional?

Answer

Yes, provided the specific manufacturer conditions are met, such as device and lead model compatibility, programming to an MRI-safe mode, and appropriate field strength limits.

#358 · Devices

What is the pocket infection risk difference between de novo device implantation and generator replacement?

Answer

Generator replacement carries a higher infection risk than initial implantation, largely due to re-entry into a previously colonized or scarred pocket and longer procedure time.

#359 • Devices

What organism most commonly causes early CIED pocket infection?

Answer

Staphylococcus species, particularly Staphylococcus aureus and coagulase-negative staphylococci such as Staphylococcus epidermidis, account for the majority of early infections.

#360 • Devices

What is the recommended treatment approach for confirmed CIED systemic infection or lead-associated endocarditis?

Answer

Complete hardware removal, including the generator and all leads, combined with pathogen-directed antibiotic therapy, since antibiotics alone rarely eradicate infection on hardware.

#361 • Devices

What is the biophysical basis for how bipolar sensing reduces far-field oversensing compared to unipolar sensing?

Answer

Bipolar sensing uses two closely spaced electrodes on the same lead, creating a small sensing vector that captures only local myocardial signal and rejects distant electrical noise.

#362 • Devices

What is impedance in the context of a pacing or defibrillation lead, and why is it monitored?

Answer

Impedance is the resistance to current flow through the lead-tissue circuit, and trending it detects lead problems such as insulation breaches (low impedance) or conductor fractures (high impedance).

#363 · Devices

What automated external defibrillator (AED) feature allows it to be used safely by laypersons?

Answer

The AED automatically analyzes the rhythm and only permits shock delivery when it detects a shockable rhythm such as ventricular fibrillation or pulseless ventricular tachycardia.

#364 · Devices

What is the main advantage of a wearable cardioverter-defibrillator (WCD) over an implanted ICD in the early post-MI period?

Answer

It provides temporary protection against sudden cardiac death during a period when ejection fraction may recover and permanent ICD implantation may not yet be indicated.

#365 • Devices

What is a key limitation of the wearable cardioverter-defibrillator compared to an ICD?

Answer

It depends on patient compliance with continuous wear and functioning skin electrodes, so it cannot treat arrhythmias that occur while the device is off or improperly worn.

#366 • Devices

What is the class I indication for permanent pacing in symptomatic sinus node dysfunction?

Answer

Permanent pacing is indicated when there is documented correlation between symptoms such as syncope or fatigue and bradycardia or sinus pauses attributable to sinus node dysfunction.

#367 · Devices

What is the class I indication for permanent pacing in acquired third-degree AV block?

Answer

Permanent pacing is indicated regardless of symptoms, because complete heart block carries a substantial risk of syncope, heart failure, or sudden death from an unreliable escape rhythm.

#368 · Devices

How does Mobitz type II second-degree AV block differ from Mobitz type I in terms of pacing indication?

Answer

Mobitz type II reflects infra-Hisian disease with risk of sudden progression to complete heart block and generally warrants pacing, while Mobitz type I (Wenckebach) is usually benign and often does not.

#369 · Devices

What is a common cause of pacemaker syndrome, and how does it typically present?

Answer

It occurs mainly with single-chamber ventricular pacing causing AV dyssynchrony, presenting with fatigue, dyspnea, and hypotension due to loss of atrial contribution to ventricular filling and atrial contraction against a closed AV valve.

#370 · Devices

How is pacemaker syndrome typically managed once diagnosed?

Answer

Upgrade to dual-chamber pacing to restore AV synchrony, which resolves the hemodynamic and symptomatic consequences of ventriculoatrial conduction and loss of atrial kick.

#371 · Devices

What is the target implant location for a standard right ventricular pacing lead to minimize dyssynchrony compared to the RV apex?

Answer

The right ventricular septum, particularly the mid-septal or septal outflow tract region, is preferred over the apex because it produces a more physiologic activation pattern and less mechanical dyssynchrony.

#372 · Devices

What complication can result from chronic high-burden right ventricular apical pacing?

Answer

Pacing-induced cardiomyopathy, from electrical and mechanical dyssynchrony that mimics a left bundle branch block pattern, can progress to reduced ejection fraction and heart failure over time.

#373 · Devices

What is the mechanism of a pneumothorax as a complication of transvenous lead implantation?

Answer

Inadvertent puncture of the pleura during subclavian or axillary venous access allows air to enter the pleural space, which can cause lung collapse.

#374 · Devices

What is cardiac tamponade as an implant complication, and how does it present?

Answer

It results from lead perforation of the myocardium allowing blood to accumulate in the pericardial space, presenting with hypotension, distended neck veins, and muffled heart sounds (Beck's triad).

#375 · Devices

What programming parameter defines the lowest rate at which a pacemaker will pace in the absence of intrinsic activity?

Answer

The lower rate limit (base rate), below which the device will deliver a pacing stimulus if no intrinsic beat is sensed within the escape interval.

#376 · Devices

What is the purpose of the pacemaker's upper rate limit in dual-chamber pacing?

Answer

It caps the ventricular tracking rate in response to sensed atrial events, preventing excessively rapid ventricular pacing during fast atrial rates such as sinus tachycardia or atrial arrhythmias.

#377 · Devices

What is pacemaker-mediated tachycardia (PMT), and what triggers it?

Answer

PMT is an endless-loop tachycardia in dual-chamber pacemakers triggered when a ventricular event conducts retrogradely to the atrium, which is sensed and tracked, driving repetitive ventricular pacing at the upper rate limit.

#378 · Devices

How is pacemaker-mediated tachycardia typically terminated or prevented programmatically?

Answer

Extending the postventricular atrial refractory period (PVARP) prevents the device from sensing the retrograde P wave, and most devices have an automatic PMT-termination algorithm that extends PVARP after one cycle.

#379 · Devices

What is the general depth of implantation of a single-chamber leadless pacemaker such as the Micra?

Answer

It is delivered transvenously via the femoral vein and implanted directly in the right ventricular myocardium using tines or a helix fixation, with no lead or subcutaneous pocket.

#380 · Devices

What is the main clinical advantage of a leadless pacemaker over a conventional transvenous system?

Answer

Elimination of the transvenous lead and subcutaneous pocket removes the two most common sources of long-term complications, namely lead fracture/failure and pocket infection.

#381 • Devices

What is a key limitation of the Micra AV single-chamber leadless pacemaker's atrial sensing?

Answer

The Micra AV senses atrial contraction indirectly using an internal accelerometer to detect the mechanical atrial kick, rather than sensing a true atrial electrical signal, so AV synchrony can be less reliable than with a true dual-chamber system, especially with an irregular atrial rhythm or poor patient positioning.

#382 • Devices

What is a limitation of accelerometer-based atrial sensing in the single-chamber Micra AV leadless pacemaker?

Answer

Its accuracy can be reduced by patient motion, atrial arrhythmias, or weak atrial contraction, which can degrade AV synchrony compared to a true dual-chamber system using electrical atrial sensing. True dual-chamber leadless systems instead use implant-to-implant conducted electrical communication between two separate capsules, not accelerometer-based sensing.

#383 · Devices

What is a unique retrieval consideration for leadless pacemakers compared to transvenous leads?

Answer

Leadless devices are generally designed to be retrievable with a dedicated snare system for a period after implant, but chronic fibrous encapsulation can make retrieval difficult or impossible after several years, so most are left in place and a new device is added when the battery depletes.

#384 · Devices

What is the target site for His bundle pacing?

Answer

The lead is fixed at or near the His bundle within the membranous septum, aiming to capture the native conduction system directly and preserve physiologic ventricular activation.

#385 • Devices

What is a key technical challenge specific to His bundle pacing?

Answer

Higher and less stable pacing thresholds and lower sensed R-wave amplitudes compared to conventional pacing, along with technically demanding lead placement and risk of dislodgement.

#386 • Devices

What is left bundle branch area pacing (LBBAP), and how does it differ anatomically from His bundle pacing?

Answer

LBBAP involves deep septal lead placement that captures the left bundle branch fascicles beyond the point of block, positioned deeper in the interventricular septum than the His bundle, giving more stable thresholds and wider correction of conduction disease.

#387 · Devices

What is a major clinical advantage of LBBAP over His bundle pacing in practice?

Answer

LBBAP generally achieves lower and more stable capture thresholds with higher R-wave amplitudes, and can correct distal conduction disease that His bundle pacing cannot reliably capture.

#388 · Devices

What ECG finding suggests successful left bundle branch capture during LBBAP implantation?

Answer

A right bundle branch block pattern in lead V1 with a short and stable stimulus-to-left ventricular activation time (measured to the R-wave peak in V5/V6) supports conduction system capture.

#389 · Devices

What is the class I indication for cardiac resynchronization therapy (CRT) in terms of ejection fraction, QRS morphology, and duration?

Answer

CRT is class I indicated for patients with LVEF 35% or less, left bundle branch block morphology, QRS duration 150 ms or greater, and NYHA class II-IV symptoms on guideline-directed medical therapy.

#390 · Devices

Why do patients with left bundle branch block morphology respond better to CRT than those with non-LBBB QRS morphology?

Answer

LBBB produces a more predictable pattern of delayed lateral wall activation that CRT's biventricular pacing can effectively correct, whereas non-LBBB morphologies (such as intraventricular conduction delay) reflect more heterogeneous dyssynchrony that resynchronization corrects less reliably.

#391 · Devices

What is the target vessel for placement of the left ventricular lead during standard CRT implantation?

Answer

A tributary of the coronary sinus, typically a lateral or posterolateral branch, is cannulated to position the LV lead epicardially over the left ventricular free wall.

#392 · Devices

What complication is specific to CRT implantation related to the coronary sinus lead?

Answer

Coronary sinus dissection or perforation during venography or lead advancement, which can cause pericardial effusion or tamponade.

#393 · Devices

What is phrenic nerve stimulation as a CRT complication, and how is it typically managed?

Answer

It occurs when the LV lead's pacing vector captures the adjacent phrenic nerve, causing diaphragmatic twitching; management includes reprogramming to an alternate pacing vector or repositioning the lead if reprogramming fails.

#394 · Devices

What is CRT non-response, and what percentage of patients approximately experience it?

Answer

CRT non-response is failure to achieve meaningful clinical or echocardiographic improvement despite device implantation, occurring in roughly 30% of patients.

#395 • Devices

What is one of the most common reversible causes of CRT non-response?

Answer

Suboptimal biventricular pacing percentage, often due to frequent premature ventricular contractions or atrial fibrillation with rapid conduction that prevents consistent CRT delivery, requiring a pacing percentage above about 98% for benefit.

#396 • Devices

How can AV and VV interval optimization address CRT non-response?

Answer

Adjusting the atrioventricular delay optimizes the timing of ventricular filling relative to atrial contraction, and adjusting the interventricular (VV) delay fine-tunes the relative timing of right and left ventricular activation to maximize resynchronization.

#397 • Devices

What is the class I indication for ICD implantation for primary prevention in ischemic cardiomyopathy?

Answer

LVEF 35% or less, NYHA class II-III symptoms despite at least 3 months of guideline-directed medical therapy, and at least 40 days post-myocardial infarction, with reasonable expectation of survival beyond one year.

#398 • Devices

Why must ICD implantation for primary prevention after myocardial infarction wait at least 40 days?

Answer

Ejection fraction can recover substantially in the early post-MI period, and trial data (such as DINAMIT) showed no mortality benefit from very early ICD implantation despite reduced arrhythmic death, offset by increased non-arrhythmic death.

#399 · Devices

What defines a secondary prevention indication for ICD implantation?

Answer

Prior sustained ventricular tachycardia or ventricular fibrillation causing hemodynamic instability or cardiac arrest, not due to a completely reversible cause, warrants ICD implantation regardless of ejection fraction.

#400 · Devices

What is the purpose of programming a detection duration (number of intervals to detect) before ICD therapy delivery?

Answer

It requires the fast rhythm to persist for a set number of consecutive beats before therapy is delivered, reducing inappropriate shocks for self-terminating nonsustained arrhythmias.

#401 • Devices

What is the evidence-based rationale for programming higher detection rate cutoffs and longer detection durations in primary prevention ICD patients (per MADIT-RIT)?

Answer

High-rate, long-duration detection programming significantly reduces inappropriate shocks and all-cause mortality compared to conventional programming, without increasing the risk of syncope or missed appropriate therapy.

#402 • Devices

What is antitachycardia pacing (ATP), and why is it preferred as a first-line therapy for monomorphic VT?

Answer

ATP delivers a burst of rapid pacing to interrupt a reentrant VT circuit painlessly, and it terminates a substantial proportion of monomorphic VT episodes without requiring a painful shock.

#403 · Devices

What is a lead-related complication unique to ICD leads compared to standard pacing leads?

Answer

Lead fracture or insulation failure can cause oversensing of noise, which the device may misinterpret as ventricular fibrillation, resulting in inappropriate shocks.

#404 · Devices

What is T-wave oversensing, and why is it a concern in ICD programming?

Answer

T-wave oversensing occurs when the device's sensing algorithm counts the T wave as an additional ventricular event, effectively double-counting the heart rate and potentially triggering inappropriate detection of a fast rhythm.

#405 · Devices

How does the subcutaneous ICD (S-ICD) differ anatomically from a transvenous ICD?

Answer

The S-ICD lead and generator are placed entirely outside the vasculature and heart, with the lead tunneled parasternally in the subcutaneous tissue and the generator implanted laterally, avoiding any intravascular or intracardiac hardware.

#406 · Devices

What is the main advantage of the S-ICD regarding long-term vascular and cardiac complications?

Answer

It eliminates the risks of transvenous lead-related complications such as lead fracture, venous thrombosis or occlusion, tricuspid valve damage, and endovascular infection or endocarditis.

#407 · Devices

What is a key functional limitation of the S-ICD compared to a transvenous ICD?

Answer

It cannot provide antitachycardia pacing or chronic bradycardia pacing, since it has no lead in contact with the myocardium, so it delivers only shock therapy for detected arrhythmias.

#408 · Devices

Why is preimplant ECG screening required before S-ICD implantation?

Answer

Screening in multiple postures and vectors identifies patients whose QRS/T-wave morphology risks T-wave oversensing or inadequate R-wave amplitude, which could cause inappropriate shocks or failed detection with the device's surface-sensed signal.

#409 • Devices

In which patients is the S-ICD generally preferred over a transvenous ICD?

Answer

Patients without a pacing indication, particularly younger patients, those with difficult venous access, prior device infection, or high infection risk, since the S-ICD avoids intravascular hardware.

#410 • Devices

What is a contraindication to S-ICD implantation related to bradycardia therapy needs?

Answer

Patients who require chronic bradycardia pacing, cardiac resynchronization therapy, or antitachycardia pacing for monomorphic VT are not appropriate candidates, since the S-ICD cannot deliver any of these therapies.

#411 · Devices

What is the extravascular ICD (EV-ICD), and how does its lead placement differ from both transvenous and subcutaneous systems?

Answer

The EV-ICD places its lead substernally in the anterior mediastinal space rather than intravascularly or purely subcutaneously, allowing it to deliver ATP and lower-energy defibrillation while still avoiding intravascular and intracardiac hardware.

#412 · Devices

What functional capability does the extravascular ICD offer that the subcutaneous ICD lacks?

Answer

The EV-ICD can deliver antitachycardia pacing and backup bradycardia pacing because its substernal lead position allows for pacing capability, unlike the S-ICD.

#413 • Devices

What is the primary indication for lead extraction related to infection?

Answer

Any evidence of systemic CIED infection, including bacteremia with no other clear source in a patient with hardware, valvular or lead vegetation/endocarditis, or pocket infection, is a class I indication for complete hardware removal.

#414 • Devices

Besides infection, what other clinical scenario is a class I indication for lead extraction?

Answer

Life-threatening arrhythmia caused by a retained or fractured lead, or a lead that poses immediate patient risk such as significant thrombus formation causing symptomatic obstruction, warrants extraction.

#415 · Devices

Why does lead extraction become progressively more difficult and risky as lead dwell time increases?

Answer

Fibrous tissue and calcification progressively encase the lead within the vein and cardiac structures over time, increasing adhesion to vascular and myocardial tissue and raising the risk of vessel or cardiac perforation during extraction.

#416 · Devices

What is the most catastrophic complication of lead extraction, and why is surgical backup required?

Answer

Superior vena cava or cardiac perforation causing acute tamponade or hemorrhage is the most feared complication, so extraction of chronically implanted leads should be performed with immediate cardiothoracic surgical backup and pericardiocentesis capability available.

#417 • Devices

What extraction tools are commonly used to free a chronically fibrosed lead from the vasculature?

Answer

Mechanical sheaths (including laser or radiofrequency-powered sheaths) are advanced over the lead to dissect fibrous adhesions, allowing controlled countertraction removal of the encased lead.

#418 • Devices

What is the recommended timing for reimplantation of a new device after extraction for infection?

Answer

Reimplantation is generally delayed until the infection has been adequately treated with a completed antibiotic course and blood cultures have cleared, with timing individualized based on the presence of bacteremia, endocarditis, or vegetation.

#419 · Devices

What is the risk of implanting a new device on the contralateral side immediately after extraction for a still-active infection?

Answer

Residual bacteremia or incompletely treated infection can seed the new hardware, leading to recurrent device infection even at a completely new site.

#420 · Devices

What is an insertable (implantable) loop recorder (ILR), and what is its primary clinical use?

Answer

It is a small subcutaneous device that continuously monitors and stores cardiac rhythm data for extended periods, primarily used to investigate unexplained syncope or to detect infrequent arrhythmias such as paroxysmal atrial fibrillation.

#421 • Devices

What is the typical battery longevity of a modern insertable loop recorder?

Answer

Approximately 3 to 4.5 years, depending on the manufacturer and monitoring settings.

#422 • Devices

Why is an ILR often preferred over a 30-day event monitor for evaluating infrequent syncope?

Answer

Its multi-year monitoring duration greatly increases the likelihood of capturing an infrequent arrhythmic event that a short-term external monitor would likely miss.

#423 • Devices

How does an insertable loop recorder detect atrial fibrillation automatically?

Answer

It uses an algorithm that analyzes RR interval irregularity to flag episodes suggestive of atrial fibrillation, which are then stored and typically require physician review to confirm.

#424 • Devices

What is a recognized limitation of automated AF detection algorithms in insertable loop recorders?

Answer

False-positive detections can occur due to irregular sensing from ectopy, undersensing, or noise, so recorded episodes generally require manual adjudication rather than automatic acceptance.

#425 · Devices

What is the general concept behind AI-based remote monitoring triage systems for CIED data?

Answer

Machine learning algorithms analyze large volumes of remotely transmitted device data (arrhythmia burden, impedance trends, heart failure diagnostics) to prioritize and flag clinically significant alerts, reducing the burden of manual review of routine transmissions.

#426 · Devices

What is a key clinical benefit of AI-based triage in remote CIED monitoring programs?

Answer

It can help clinical staff efficiently identify actionable findings, such as impending lead failure or rising heart failure risk scores, among a high volume of transmissions, potentially reducing time to intervention.

#427 · Devices

What is a recognized limitation or concern regarding AI-based triage algorithms in device monitoring?

Answer

Algorithm performance depends heavily on training data quality and may not generalize equally across all patient populations or device platforms, and clinical oversight remains necessary since false negatives could delay recognition of a true clinical event.

#428 · Devices

What multi-sensor heart failure diagnostic index combines several device-measured parameters to predict decompensation risk?

Answer

Composite heart failure risk scores integrate parameters such as intrathoracic impedance, activity level, heart rate variability, and arrhythmia burden to generate a risk index that flags patients for closer clinical follow-up.

#429 · Devices

What is the difference between a passive fixation lead and an active fixation lead?

Answer

Passive fixation leads use tines that lodge in the trabeculae of the heart to anchor position, while active fixation leads use a screw-in helix that engages directly into the myocardium, allowing more flexible lead positioning.

#430 · Devices

Why might an active fixation lead be preferred over a passive fixation lead?

Answer

Active fixation allows placement in locations without prominent trabeculae, such as the right ventricular septum or right atrial appendage alternatives, and generally allows for easier extraction if needed.

#431 · Devices

What is the significance of an elevated pacing threshold at follow-up in a patient with a previously stable lead?

Answer

It may indicate lead maturation issues, microdislodgement, fibrosis at the tip, exit block, or steroid elution depletion, and can risk loss of capture if it approaches the programmed output.

#432 · Devices

What is the purpose of steroid-eluting lead tips in modern pacing and defibrillation leads?

Answer

The steroid reduces local inflammation and fibrosis at the electrode-tissue interface, helping keep chronic pacing thresholds low and stable compared to older non-steroid leads.

#433 · Devices

What is twiddler's syndrome, and how does it present?

Answer

It occurs when a patient manipulates the device generator within its pocket, causing the leads to coil and dislodge, presenting with loss of capture, diaphragmatic or extracardiac stimulation, or device malfunction.

#434 · Devices

What patient population is at higher risk for twiddler's syndrome?

Answer

Patients with cognitive impairment, obesity with a loose subcutaneous pocket, or a habit of manipulating the device site are at increased risk.

#435 • Devices

What is the mechanism of subclavian crush syndrome affecting a pacing or ICD lead?

Answer

Lead placement between the clavicle and first rib via a medial subclavian venous approach can cause chronic mechanical compression, leading to insulation damage or conductor fracture over time.

#436 • Devices

Why is the axillary or cephalic vein approach preferred over the medial subclavian approach for lead placement?

Answer

It avoids the costoclavicular space where lead compression (subclavian crush) can occur, reducing long-term risk of lead insulation or conductor damage, and also lowers pneumothorax risk.

#437 · Devices

What defines undersensing on a device interrogation, and what is a clinical consequence?

Answer

Undersensing is failure of the device to appropriately detect intrinsic cardiac signals, which can result in inappropriate pacing that competes with the native rhythm, potentially causing R-on-T phenomenon and triggering ventricular arrhythmia.

#438 · Devices

What defines oversensing on a device interrogation, and what is a clinical consequence in a pacemaker-dependent patient?

Answer

Oversensing is inappropriate detection of non-cardiac or extraneous signals (myopotentials, EMI, lead noise) as intrinsic beats, which can cause inappropriate inhibition of pacing output and resultant symptomatic bradycardia or asystole.

#439 · Devices

What is crosstalk in a dual-chamber pacemaker, and how is it prevented?

Answer

Crosstalk occurs when an atrial pacing output is sensed by the ventricular channel and misinterpreted as an intrinsic ventricular event, inappropriately inhibiting ventricular pacing; it is prevented using a ventricular blanking period after atrial output and ventricular safety pacing algorithms.

#440 · Devices

What is ventricular safety pacing, and what does it protect against?

Answer

It is a short algorithm window after atrial pacing during which any sensed ventricular signal triggers an immediate safety ventricular pace at a short AV delay, protecting against the consequence of crosstalk causing missed ventricular pacing.

#441 • Devices

What is the purpose of rate-responsive (rate-adaptive) pacing, and what sensors commonly enable it?

Answer

Rate-responsive pacing allows the paced rate to increase with physical activity to mimic normal chronotropic response, most commonly using an accelerometer sensor that detects body motion.

#442 • Devices

In what patient population is rate-responsive pacing particularly beneficial?

Answer

Patients with chronotropic incompetence, who cannot appropriately increase their intrinsic heart rate with exertion, benefit most from rate-adaptive pacing to improve exercise capacity.

#443 • Devices

What is mode switching in a dual-chamber pacemaker, and when does it activate?

Answer

Mode switching is an automatic algorithm that changes the device from an atrial-tracking mode to a non-tracking mode when a rapid atrial arrhythmia such as atrial fibrillation is detected, preventing inappropriate rapid ventricular tracking of the arrhythmia.

#444 • Devices

What is far-field R-wave oversensing in the atrial channel, and why is it clinically relevant?

Answer

It occurs when the atrial lead inappropriately senses the ventricular depolarization signal as an atrial event, which can trigger inappropriate mode switching or misdiagnosis of atrial arrhythmia on remote monitoring.

#445 · Devices

What is the significance of high defibrillation threshold (DFT) at ICD implant testing?

Answer

A high DFT indicates that a greater energy delivery is required to successfully terminate induced ventricular fibrillation, which may prompt repositioning of the shocking coil, adding a subcutaneous array, or reversing shock polarity to improve the safety margin.

#446 · Devices

Why has routine defibrillation threshold testing at implant become less common in current practice?

Answer

Studies have shown that omitting routine DFT testing does not increase the rate of failed first shocks or arrhythmic death, while avoiding the risks associated with inducing ventricular fibrillation during the procedure.

#447 · Devices

What is the purpose of a shocking coil integrated into a standard transvenous ICD lead?

Answer

The coil, typically positioned in the right ventricle (and sometimes also the superior vena cava), delivers the high-voltage defibrillation shock directly within the vasculature near the heart to terminate ventricular fibrillation or unresponsive VT.

#448 · Devices

What is the difference between single-coil and dual-coil ICD leads?

Answer

Single-coil leads have a shocking coil only in the right ventricle, while dual-coil leads add a second coil in the superior vena cava; dual-coil leads may lower defibrillation thresholds slightly but are generally more difficult to extract.

#449 · Devices

What is lead noise/make-break potential, and how does it typically present on remote monitoring?

Answer

It results from intermittent conductor fracture or connector looseness causing erratic, nonphysiologic high-frequency signals on impedance and EGM tracings, which can trigger inappropriate shocks from misclassification as ventricular fibrillation.

#450 · Devices

What is the general infection risk difference between an S-ICD and a transvenous ICD system?

Answer

The S-ICD has a comparable or somewhat higher rate of pocket-related infection due to larger generator size and subcutaneous positioning, but any infection is confined outside the vascular and cardiac space, avoiding the risk of endovascular seeding or endocarditis.

#451 · Devices

What is CIED-associated superior vena cava (SVC) syndrome, and what causes it?

Answer

It results from chronic venous obstruction or thrombosis around multiple leads within the SVC, presenting with facial and upper extremity swelling, and is caused by lead-related endothelial injury and fibrotic narrowing over time.

#452 · Devices

What is the purpose of a diagnostic His bundle electrogram during electrophysiology study prior to considering conduction system pacing?

Answer

It confirms the location of the His bundle and characterizes conduction (AH and HV intervals), helping guide precise lead placement for His bundle pacing and assess baseline conduction disease.

#453 · Devices

What is a septal perforation as a complication specific to conduction system pacing (His bundle or left bundle branch area pacing)?

Answer

The active fixation lead can be advanced too far through the interventricular septum into the left ventricular cavity or beyond, risking mitral valve chordal injury, ventricular perforation, or embolic complications from an intracavitary lead tip.

#454 · Devices

How is septal lead depth typically confirmed during LBBAP implantation to avoid overshooting into the LV cavity?

Answer

Using paced QRS morphology transition (unipolar tip current-of-injury pattern), fluoroscopic lead tip position across the septum, and impedance drop patterns during screwing, combined with intraprocedural pacing morphology assessment.

#455 · Devices

What is the significance of a QS pattern versus an R-wave pattern in the unipolar pacing electrogram during LBBAP lead deployment?

Answer

A transition from an R-wave (lead tip still within the septum) to a QS pattern signals that the lead tip has crossed into the left ventricular subendocardium, confirming adequate depth for left bundle capture.

#456 · Devices

What is the general recommendation regarding CRT candidacy in patients with permanent atrial fibrillation and an indication for AV nodal ablation?

Answer

Patients with permanent AF, reduced ejection fraction, and a need for rate control via AV nodal ablation are often considered for CRT (the pace-and-ablate strategy) to ensure near-100% biventricular pacing rather than standard right ventricular pacing alone.

#457 · Devices

What is the concern with programming an ICD's SVC coil to be electrically active in a patient found to have chronic SVC thrombosis or occlusion?

Answer

An occluded or thrombosed SVC can alter the shocking vector and increase defibrillation impedance, potentially raising the defibrillation threshold and reducing shock efficacy, so reprogramming to exclude the SVC coil from the shocking circuit may be needed.

#458 · Devices

What is the purpose of a device's stored intracardiac electrogram (EGM) during a detected arrhythmia episode?

Answer

It allows retrospective review of the actual sensed signal morphology and timing during an event, helping distinguish a true ventricular arrhythmia from oversensing, lead artifact, or supraventricular tachycardia with rapid conduction.

#459 · Devices

Why might discrimination algorithms based on EGM morphology or onset/stability criteria reduce inappropriate ICD shocks?

Answer

They help distinguish ventricular tachyarrhythmias from supraventricular tachycardias with rapid ventricular rates by analyzing features such as gradual versus abrupt rate onset, rate stability, and QRS morphology similarity to the patient's baseline rhythm.

#460 · Devices

What is a key long-term surveillance consideration for patients with a leadless pacemaker regarding future device management?

Answer

Since batteries eventually deplete and retrieval may become difficult after chronic implantation, planning must consider whether a second leadless device can be added alongside the depleted one or whether the existing device should be left in place and deactivated.

#461 · Clinical Scenarios and Syndromes

What historical and exam features distinguish cardiac syncope from vasovagal syncope?

Answer

Cardiac syncope typically occurs suddenly with little or no prodrome, may happen during exertion or while supine, and is associated with structural heart disease, palpitations, or a family history of sudden death. Vasovagal syncope usually has a prodrome of warmth, nausea, or diaphoresis and is triggered by prolonged standing, pain, or emotional stress.

#462 · Clinical Scenarios and Syndromes

In an unexplained syncope patient with normal cardiac workup, what is the role of an implantable loop recorder (ILR)?

Answer

An ILR provides long term continuous monitoring (up to 3 years) and is recommended when syncope is infrequent or unexplained after initial evaluation, since it substantially increases diagnostic yield compared to conventional short term monitoring.

#463 · Clinical Scenarios and Syndromes

What is the first step in evaluating a patient presenting with palpitations?

Answer

A detailed history characterizing the onset, offset, rate, regularity, and associated symptoms, combined with a 12 lead ECG during an episode if possible, since the differential and workup depend heavily on whether the rhythm is regular or irregular and fast or slow.

#464 · Clinical Scenarios and Syndromes

A patient describes palpitations with a sensation of neck pounding along with the heartbeat. What does this suggest?

Answer

Regular cannon A waves (the 'frog sign') suggest AVNRT, caused by near-simultaneous atrial and ventricular contraction from fast retrograde conduction against a closed tricuspid valve, not AV dissociation. Irregular or intermittent cannon A waves instead suggest true AV dissociation, classically seen in ventricular tachycardia, where atrial and ventricular contractions coincide only sporadically.

#465 · Clinical Scenarios and Syndromes

What is the single strongest predictor of sudden cardiac death risk in most structural heart disease populations used for primary prevention ICD decisions?

Answer

Left ventricular ejection fraction, with an LVEF of 35% or less in ischemic or nonischemic cardiomyopathy with NYHA class II to III symptoms on optimal medical therapy generally meeting criteria for primary prevention ICD implantation.

#466 · Clinical Scenarios and Syndromes

Why should primary prevention ICD implantation typically be deferred for at least 40 days after myocardial infarction?

Answer

The DINAMIT and IRIS trials, not MADIT-II, showed no overall mortality benefit from ICD implantation in the first weeks after myocardial infarction, since a reduction in arrhythmic death was offset by an increase in non-arrhythmic death. MADIT-II instead established the benefit of primary prevention ICDs in patients with LVEF 30% or less at least one month after MI.

#467 · Clinical Scenarios and Syndromes

What ethical principle governs a competent patient's request to deactivate their ICD shock therapy at end of life?

Answer

Patient autonomy: a competent patient or their legally authorized surrogate has the right to refuse or withdraw any medical therapy, including ICD shocks, and deactivation is ethically and legally equivalent to withholding therapy, not physician-assisted suicide.

#468 · Clinical Scenarios and Syndromes

When a terminally ill patient requests ICD deactivation, should pacemaker function also be turned off?

Answer

Generally no; anti-tachycardia therapies (shocks and ATP) are typically deactivated while bradycardia pacing is continued if the patient is pacemaker dependent, since pacing supports comfort and quality of life rather than futile resuscitation from a lethal arrhythmia.

#469 · Clinical Scenarios and Syndromes

What is the physician's obligation when a device or lead advisory is issued for a device already implanted in a patient?

Answer

The physician must review the specific advisory details, assess individualized patient risk (including whether the patient is pacemaker or ICD dependent), discuss risks and benefits of monitoring versus revision with the patient, and document shared decision making.

#470 · Clinical Scenarios and Syndromes

What generally favors prophylactic extraction versus continued monitoring for a patient with a recalled lead?

Answer

Prophylactic extraction is favored when the failure rate and consequence of failure are high (such as pacemaker dependence with risk of asystole, or inability to reliably monitor for impending failure), while monitoring is favored for low risk, non-dependent patients since extraction itself carries procedural risk.

#471 · Clinical Scenarios and Syndromes

What ECG finding defines Long QT syndrome and what corrected QT interval is generally considered diagnostic in the absence of a secondary cause?

Answer

A QTc of 500 ms or greater on repeated ECGs, in the absence of a secondary cause, is diagnostic of LQTS on ECG grounds alone, as is a Schwartz score of 3.5 or higher. A QTc of 480 to 499 ms is diagnostic only when accompanied by unexplained syncope in the absence of a secondary cause.

#472 · Clinical Scenarios and Syndromes

What triggers arrhythmic events in LQT1 versus LQT2 versus LQT3?

Answer

LQT1 (KCNQ1, IKs) events are triggered by exercise, especially swimming. LQT2 (KCNH2, IKr) events are triggered by emotional stress or auditory stimuli such as an alarm. LQT3 (SCN5A, late sodium current) events often occur during sleep or at rest with bradycardia.

#473 · Clinical Scenarios and Syndromes

What is the first line therapy for all genotypes of congenital Long QT syndrome?

Answer

Beta blockers (nadolol or propranolol preferred) are first line therapy for all LQTS genotypes and reduce the risk of cardiac events, with ICD implantation reserved for those with breakthrough events on therapy or prior cardiac arrest.

#474 · Clinical Scenarios and Syndromes

What is the classic ECG pattern of Brugada syndrome Type 1 and how is it diagnosed?

Answer

Type 1 Brugada pattern shows a coved ST segment elevation of at least 2 mm followed by a negative T wave in one or more right precordial leads (V1 to V2), which may need to be recorded in a high lead position (2nd or 3rd intercostal space) or unmasked with a sodium channel blocker challenge.

#475 · Clinical Scenarios and Syndromes

What clinical circumstance classically precipitates arrhythmic events in Brugada syndrome?

Answer

Events in Brugada syndrome classically occur at rest or during sleep, often at night, and can be provoked by fever, since fever can unmask or worsen the Type 1 Brugada ECG pattern and increase arrhythmic risk.

#476 · Clinical Scenarios and Syndromes

What is the recommended management for a patient with fever and known Brugada syndrome?

Answer

Prompt and aggressive antipyretic treatment (such as acetaminophen) is recommended because fever can precipitate ventricular arrhythmias in Brugada syndrome by unmasking the Type 1 ECG pattern.

#477 · Clinical Scenarios and Syndromes

What is the characteristic exercise-induced arrhythmia pattern seen in catecholaminergic polymorphic ventricular tachycardia (CPVT)?

Answer

CPVT classically produces bidirectional ventricular tachycardia (beat to beat alternating QRS axis) or polymorphic VT that is reproducibly triggered by exercise or emotional stress in a structurally normal heart with a normal resting ECG.

#478 · Clinical Scenarios and Syndromes

What are the two main genetic causes of CPVT and their inheritance patterns?

Answer

CPVT is most commonly caused by autosomal dominant mutations in RYR2 (cardiac ryanodine receptor) and less commonly by autosomal recessive mutations in CASQ2 (calsequestrin), both leading to abnormal intracellular calcium handling.

#479 · Clinical Scenarios and Syndromes

What is first line pharmacologic therapy for CPVT, and why should sodium channel blockers or catecholamines not be relied upon alone?

Answer

Nonselective beta blockers (nadolol preferred) are first line therapy; flecainide can be added for breakthrough arrhythmias by reducing ryanodine receptor calcium leak, but beta blockade remains foundational since exercise restriction and adrenergic blockade directly target the trigger mechanism.

#480 · Clinical Scenarios and Syndromes

In hypertrophic cardiomyopathy, what wall thickness and additional risk factors are used to guide primary prevention ICD decisions?

Answer

A maximal left ventricular wall thickness of 30 mm or greater is a major risk factor; other factors include unexplained syncope, family history of sudden death, nonsustained VT, abnormal blood pressure response to exercise, and extensive late gadolinium enhancement on cardiac MRI, combined in risk calculators such as the HCM Risk-SCD score.

#481 · Clinical Scenarios and Syndromes

What is the mechanism of the dynamic left ventricular outflow tract obstruction in obstructive hypertrophic cardiomyopathy?

Answer

Systolic anterior motion of the mitral valve leaflet, pulled toward the hypertrophied septum by Venturi effect and altered flow dynamics, causes dynamic LVOT obstruction that worsens with reduced preload, reduced afterload, or increased contractility.

#482 · Clinical Scenarios and Syndromes

What ECG and imaging findings characterize arrhythmogenic right ventricular cardiomyopathy (ARVC)?

Answer

ARVC classically shows T wave inversions in the right precordial leads (V1 to V3) beyond age 14, an epsilon wave, and right ventricular dilation, aneurysm, or wall motion abnormality on imaging, reflecting fibrofatty replacement of RV myocardium.

#483 · Clinical Scenarios and Syndromes

What gene is most commonly mutated in arrhythmogenic right ventricular cardiomyopathy?

Answer

PKP2 (plakophilin-2), encoding a desmosomal protein, is the most commonly identified mutation in ARVC, consistent with its underlying pathophysiology as a disease of desmosomal dysfunction and fibrofatty myocardial replacement.

#484 · Clinical Scenarios and Syndromes

What lifestyle modification is specifically recommended for patients with ARVC, even those who are asymptomatic gene carriers?

Answer

Restriction from competitive and endurance exercise is recommended, since intense exercise accelerates disease progression, RV dilation, and arrhythmic risk in ARVC, including in asymptomatic gene-positive individuals.

#485 • Clinical Scenarios and Syndromes

Which specific dilated cardiomyopathy gene mutation confers a disproportionately high risk of sudden cardiac death and malignant ventricular arrhythmia relative to the degree of LV dysfunction?

Answer

Lamin A/C (LMNA) mutations confer high arrhythmic risk, including sudden death and ventricular arrhythmias, even with only mildly reduced LVEF, and often present with conduction system disease (AV block) preceding overt cardiomyopathy.

#486 • Clinical Scenarios and Syndromes

In a patient with LMNA cardiomyopathy, what specific risk factors favor earlier ICD implantation even with LVEF above the standard 35% threshold?

Answer

Non-missense mutation type, male sex, nonsustained VT, and LVEF less than 45% at first evaluation are risk factors that favor ICD implantation in LMNA cardiomyopathy even when LVEF does not meet standard primary prevention thresholds.

#487 · Clinical Scenarios and Syndromes

What arrhythmia and conduction findings are characteristic of cardiac sarcoidosis?

Answer

Cardiac sarcoidosis characteristically causes unexplained high-grade AV block, especially in patients under age 60, along with ventricular arrhythmias and patchy myocardial involvement seen as late gadolinium enhancement on cardiac MRI or increased uptake on FDG-PET.

#488 · Clinical Scenarios and Syndromes

In a young patient presenting with unexplained complete heart block, what diagnosis should be considered and what workup is indicated?

Answer

Cardiac sarcoidosis (or giant cell myocarditis) should be considered, and workup includes cardiac MRI with late gadolinium enhancement, FDG-PET imaging, and evaluation for extracardiac sarcoid involvement, since unexplained AV block in a younger patient is atypical for age related fibrosis alone.

#489 · Clinical Scenarios and Syndromes

What ECG finding defines Short QT Syndrome?

Answer

A QTc of 360 ms or less (with values under 320 ms considered highly specific), often accompanied by tall, peaked, symmetric T waves and a short or absent ST segment, associated with risk of atrial and ventricular fibrillation and sudden death.

#490 · Clinical Scenarios and Syndromes

What ECG pattern defines early repolarization syndrome and when is it considered a malignant risk marker?

Answer

Early repolarization is defined by J point elevation with QRS slurring or notching of at least 0.1 mV in at least two contiguous inferior or lateral leads; it becomes a malignant risk marker when associated with a horizontal or descending ST segment after the J wave, especially in a patient resuscitated from unexplained VF.

#491 · Clinical Scenarios and Syndromes

What is the most common arrhythmia to occur or worsen during pregnancy, and why does pregnancy increase arrhythmia risk generally?

Answer

Supraventricular tachycardia, particularly AVNRT, is the most common arrhythmia in pregnancy; increased arrhythmia risk relates to hemodynamic changes (increased blood volume and heart rate), hormonal effects, and increased sympathetic tone.

#492 · Clinical Scenarios and Syndromes

Which antiarrhythmic drugs are generally considered safest for use during pregnancy for maternal SVT or ventricular arrhythmias?

Answer

Beta blockers (metoprolol or propranolol) and digoxin are generally considered safe; sotalol and flecainide may be used with caution when needed, while amiodarone is avoided except in life threatening refractory arrhythmias due to fetal thyroid toxicity and other adverse effects.

#493 · Clinical Scenarios and Syndromes

Is electrical cardioversion safe during pregnancy for a hemodynamically unstable arrhythmia?

Answer

Yes, electrical cardioversion is considered safe at all stages of pregnancy since the amount of current reaching the fetus is minimal, and it should not be withheld in a hemodynamically unstable mother, with fetal heart rate monitoring recommended when feasible.

#494 · Clinical Scenarios and Syndromes

Why can highly trained endurance athletes develop sinus bradycardia, first degree AV block, or Mobitz Type I block as normal findings?

Answer

Increased vagal tone from chronic athletic conditioning produces these findings, which are considered normal physiologic adaptations in asymptomatic athletes and typically resolve or improve with exercise, distinguishing them from pathologic conduction disease.

#495 · Clinical Scenarios and Syndromes

What ECG or clinical findings in an athlete should prompt further evaluation rather than being attributed to athletic conditioning?

Answer

Findings such as T wave inversion beyond V1 to V2 (especially with associated symptoms), pathologic Q waves, a Brugada pattern, marked QRS widening, or an unusually short or prolonged QTc should prompt further evaluation for underlying structural or inherited arrhythmia syndromes rather than being attributed to training alone.

#496 · Clinical Scenarios and Syndromes

What is the most common cause of sudden cardiac death in young competitive athletes in the United States?

Answer

Hypertrophic cardiomyopathy is the most common structural cause of sudden cardiac death in young competitive athletes in the United States, followed by coronary artery anomalies and other structural and electrical causes.

#497 • Clinical Scenarios and Syndromes

What is the most common late arrhythmia complication after surgical repair of tetralogy of Fallot?

Answer

Sustained monomorphic ventricular tachycardia related to macroreentry around anatomic and surgical scar (such as the ventriculotomy or VSD patch) is the most common late arrhythmia, along with risk of sudden cardiac death, particularly with a QRS duration of 180 ms or greater.

#498 • Clinical Scenarios and Syndromes

What arrhythmia is most commonly seen late after the Mustard or Senning atrial switch procedure for transposition of the great arteries?

Answer

Intra-atrial reentrant tachycardia (atrial flutter-like) and sinus node dysfunction are most common late after atrial switch procedures, related to extensive atrial suture lines and baffle scarring, along with risk of sudden death from the systemic right ventricle failing.

#499 · Clinical Scenarios and Syndromes

Why is transvenous lead placement often avoided or modified in adult congenital heart disease patients with intracardiac shunts?

Answer

Transvenous leads crossing an intracardiac shunt (such as an unrepaired ASD or after a Fontan procedure) risk systemic thromboembolism and paradoxical embolic stroke, so epicardial lead placement or careful shunt closure planning is often preferred.

#500 · Clinical Scenarios and Syndromes

What is the arrhythmia substrate most characteristic of a Fontan circulation, and why is it particularly difficult to manage?

Answer

Intra-atrial reentrant tachycardia is most characteristic of the Fontan circulation, and it is difficult to manage because extensive atrial scarring and surgical baffles create complex, often multiple reentrant circuits, while the single ventricle physiology makes hemodynamic tolerance of arrhythmia and catheter access for ablation more challenging.